UNIVERSIDAD AUTÓNOMA DE NUEVO LEÓN

FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA

SUBDIRECCIÓN DE ESTUDIOS DE POSGRADO

YIELD-AWARE HIGHER-ORDER TRAJECTORY OPTIMIZATION FOR STEP-AND-SCAN WAFER STAGES

POR
ROBERTO TREVIÑO CERVANTES

COMO REQUISITO PARA OBTENER EL GRADO DE
MAESTRÍA EN CIENCIAS DE LA INGENIERÍA ELÉCTRICA

OCTUBRE 2026

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Los miembros del Comite de Tesis recomendamos que la tesis "**Yield-Aware Higher-Order Trajectory Optimization for Step-and-Scan Wafer Stages**" realizada por el(la) alumno(a) **Roberto Treviño Cervantes**, con numero de matricula 1915003, sea aceptada para su defensa como opcion al grado de **Maestría en Ciencias de la Ingenieria Electrica.**

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A ... por su apoyo ... ETC

Agradecimientos

Agradezco a ... blah blah blah.

Abstract

Step-and-scan photolithography scanners expose the layers of an integrated circuit by sweeping a reticle and a silicon wafer through a projection slit in opposite directions, synchronized to nanometre-level accuracy. The reference trajectories that command these motions determine how much energy is injected into the poorly damped structural modes of the machine, and therefore how large the residual synchronization error is during exposure. Trajectory design is conventionally evaluated through servo-mechanical metrics — settling time, tracking error, and the moving average (MA) and moving standard deviation (MSD) of the synchronization error — none of which describes the morphology of the defects that those errors ultimately imprint on the wafer.

This work proposes a yield-aware framework for optimizing the kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\mathrm{ma}}$ of fourth-order (snap-limited) reference trajectories. A planar multi-body model of a generic lithography machine, comprising the reticle, optics, and wafer chains coupled through Kelvin–Voigt elements, is implemented as a simulation in the MuJoCo physics engine and validated against the characterization study it derives from. The synchronization error produced during scanning is transformed into a simulated wafer defect map, which a convolutional neural network (CNN) trained on the industrial WM-811K dataset (172,950 labelled wafer maps, 96.88 % hold-out accuracy) classifies into nine defect morphologies. The probability distribution estimated

by the CNN, restricted to the classes plausibly caused by scanner motion, defines a cost function that links trajectory parameters directly to expected yield, closing the loop for data-driven trajectory optimization under a throughput constraint. A recurrent surrogate model trained on randomized simulation runs accelerates the evaluation of candidate trajectories during the search.

Keywords: photolithography, step-and-scan, trajectory generation, fourth-order profiles, multi-body dynamics, convolutional neural networks, wafer maps, yield.

Symbols and notation

$\dot{x}, \ddot{x}$	Dots denote derivatives with respect to time, e.g. $\dot{x} = dx/dt$.
x^T, A^T	Transpose of the vector x and the matrix A.
$\mathbb{R}$	The field of real numbers.
$\|x\|$	A norm of x.
$\mathcal{F}_0, \mathcal{F}_1$	World (inertial) frame and base frame of the machine.
j	Chain index: $j = 1$ reticle, $j = 2$ optics, $j = 3$ wafer.
i_j	Body i within chain j; N_j is the last body of chain j.
f_{i_j}	In-plane displacement of body i_j relative to its parent body.
$\theta_{z_{i_j}}$	Rotation of body i_j about the vertical axis.
$m_{i_j}, I_{z_{i_j}}$	Mass and z-axis moment of inertia of body i_j.
K_{i_j}, C_{i_j}	Stiffness and damping of the Kelvin–Voigt coupling of body i_j.
α	Projection demagnification factor, $\alpha = 0.25$.
e_{syn}	Synchronization error between wafer and reticle stages.
$\mathrm{MA}, \mathrm{MSD}$	Moving average and moving standard deviation of e_{syn} over the exposure window.
$v_{\max}, a_{\max}$	Velocity and acceleration bounds of the reference trajectory.
$j_{\max}, s_{\max}$	Jerk and snap bounds of the reference trajectory.
CD	Critical dimension: minimum printable feature size.
NA	Numerical aperture of the projection lens.
λ	Illumination wavelength.
P_c	Probability assigned by the CNN to defect class c.
w_c	Economic weight of defect class c in the cost function.
$\mathcal{J}$	Trajectory cost: expected yield loss, $\mathcal{J} = \sum_c w_c P_c$.
WPH	Throughput in wafers per hour.
Italics	Italics denote terms in a language other than English.
Boldface	Boldface denotes technical terms at first use.

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Chapter 1

Introduction

The history of modern electronics is inseparable from the history of lithography. Since the independent invention of the integrated circuit (IC) by Jack Kilby at Texas Instruments and Robert Noyce at Fairchild Semiconductor in 1958, Moore's Law has governed the relentless progression of the semiconductor industry: the density of components on a chip doubles approximately every two years[13]. This exponential trend has translated into dramatic reductions in cost alongside equally dramatic increases in performance. The price of one gigabyte of NAND flash memory, for instance, fell from over $1000 at the start of this millennium to below $0.30 by the mid-2010s[3], while the minimum transistor feature size has shrunk from hundreds of nanometres to just a few tens of nanometres. The engine of this revolution is *photolithography* — the process by which the intricate patterns of a transistor, a wire, or a memory cell are projected onto a silicon wafer with nanometre-scale fidelity.

The fabrication of a modern IC consists of up to 30 sequential process cycles for mature technology nodes[13], each defining one distinct layer of the device; at leading-edge nodes — such as 5 nm CMOS production technology — the layer count rises to approximately 75[17;12]. A monocrystalline silicon wafer — a thin disc of up to 300 mm in diameter — serves as the substrate throughout this process. Each cycle begins with a chemical pre-treatment, such as the thermal growth of a thin oxide film for electrical isolation. A photosensitive material called *resist* is then deposited onto the wafer surface. An optical exposure system projects the pattern for the current layer — defined in a chromium film on a transparent quartz plate known as the *reticle*, or *photo-mask* — onto the resist. After development, the regions that received light are selectively dissolved (in the case of a positive resist), exposing the underlying material. This material then undergoes one of several process steps. *Plasma etching* selectively removes oxide or semiconductor material to define trenches and isolation regions. *Ion implantation* locally modifies the electrical properties of the silicon to form transistor junctions and doped wells. *Physical or chemical vapour deposition* lays down thin films of conductive metal — typically tungsten or copper — to form the interconnecting wiring that links the millions of functional elements stacked across all layers. After processing, the remaining resist is stripped and the cycle repeats for the next layer, until all layers are complete and the wafer is diced and packaged.

In the earliest days of the industry, the reticle was brought into direct physical contact with the resist-coated wafer during exposure — a technique known as *contact lithography*. While simple, this approach had two fundamental shortcomings. First, repeated physical contact progressively contaminated and damaged the delicate reticle pattern, necessitating its costly and time-consuming replacement. Second, a 1:1 projection offers no optical resolution advantage: any defect present on the reticle is transferred at full size onto the wafer. These limitations motivated a transition to *projection optical lithography*, in which a reduction lens images the reticle pattern onto the wafer from a safe working distance. The projection lens introduces a demagnification factor of four, meaning that features on the reticle are printed four times smaller on the wafer, immediately relaxing the demands on reticle fabrication. The minimum printable feature size, known as the *critical dimension* (CD), is governed by[13]:

$$\mathrm{CD} = k\,\frac{\lambda}{\mathrm{NA}}, \tag{1.0.1}$$

where λ is the illumination wavelength, NA is the numerical aperture of the projection lens, and k is a process-dependent factor. Reducing CD requires a shorter wavelength, a higher NA, and process improvements that lower k. This relation has driven a continuous push toward shorter wavelengths — from mercury arc-lamp sources at 436 nm and 365 nm, through excimer lasers at 248 nm and 193 nm, to the extreme ultraviolet (EUV) sources at 13.5 nm now entering production.

Achieving a high numerical aperture, however, comes with a practical constraint. Since $\mathrm{NA} = n \sin\vartheta$, where ϑ is the half-angle of the marginal ray accepted by the lens, attaining values of NA close to one in air requires the lens to subtend a very wide solid angle. Constructing a lens element that simultaneously covers the full area of a 300 mm wafer at high NA would require an optical element of impractical size and cost[13]. The industry's solution was to accept a smaller *exposure field* — typically 26×32 mm, corresponding to a single IC die — and expose the wafer in successive steps: after each die is exposed, the wafer stage repositions the wafer to the next die location and the exposure is repeated. This architecture defines the *wafer stepper*. In a stepper, each die is illuminated in a single, instantaneous full-field flash, with the stage brought to a complete stop before exposure commences.

While the stepper represented a decisive step forward, exposing an entire die simultaneously subjects the full field to any non-uniformities in the illumination intensity or lens aberrations. The *scanner* addresses this by replacing the full-field flash with a *scanning exposure*: the illumination system reveals only a narrow rectangular slit of the reticle, and the reticle stage sweeps the mask through this slit while the wafer stage moves synchronously in the opposite direction, maintaining the 4:1 demagnification relationship throughout[3;13]. Because the reticle moves at four times the speed of the wafer, it traverses the full reticle pattern in the same time that the wafer moves one die height. This scanning motion offers significant advantages over full-field exposure: it averages out illumination dose non-uniformities across the exposure field, improves imaging uniformity at the field edges, and allows the use of a physically larger reticle that is easier to write with electron-beam pattern generators. After each die is scanned, the wafer is stepped to the next die position and the scan repeats, defining the *step-and-scan* cycle that gives modern lithographic tools their name. A throughput of over 175 wafers per hour —

corresponding to less than 0.2 s of exposure time per die — is not uncommon in state-of-the-art tools[3].

A critical consequence of the multi-layer nature of IC fabrication is the requirement that each new layer be positioned with nanometre-level accuracy relative to all preceding layers — a specification known as *overlay*. The functional layers of a chip are interconnected by millions of tiny conductive pillars called *vias*; a misregistration between consecutive layers of even a few nanometres can misalign a via, break an intended connection, or create an unintended short circuit, rendering the IC defective[13]. Overlay requirements scale roughly in proportion with the CD and currently sit below 4 nm for leading-edge nodes. Overlay is therefore one of the primary technology drivers in scanner design, and it is governed directly by the positioning accuracy of the wafer and reticle stages during the scanning exposure.

The commercial performance of a scanner is ultimately measured in wafers per hour (WPH), since a machine whose capital cost can exceed $150 million must generate revenue continuously. Although the exposure time per die is fixed by the optics and the process, the time spent stepping and *settling* — the interval between the end of a step motion and the moment the stage has damped its residual vibrations to within the nanometre-level positioning specification — represents a significant and controllable fraction of the total cycle time[3], which places the allowable settle time below 10 ms. Every millisecond saved in settling directly translates into more dies exposed per hour and thus more chips delivered per unit time, with a lower cost per chip. Reducing settling time is therefore a central objective in scanner control engineering.

It is worth noting that, while EUV lithography at 13.5 nm has entered high-volume production for leading-edge nodes below 10 nm, deep ultraviolet (DUV) optical lithography at 193 nm remains the dominant technology for *mature nodes* — technology generations at 28 nm and above. Mature-node chips are the backbone of automotive electronics, industrial controllers, power management devices, analogue circuits, and the vast majority of consumer electronics.

The aggregate volume of wafers produced at mature nodes substantially exceeds that at leading-edge nodes, and is projected to remain so for the foreseeable future. Throughput improvements in DUV step-and-scan tools therefore continue to have a large global impact.

The key lever for reducing settling time — and thus for increasing scanner throughput — is the generation of *reference trajectories*: motion profiles that command the stage from its current position to the next scanning start-point with minimal excitation of mechanical resonances and the shortest possible residual settling tail. A well-designed trajectory limits the energy injected into the structural eigenmodes of the machine, reduces induced vibrations in the sensitive metrology and optical systems, and enables the stage to reach nanometre-level positioning accuracy well within the allocated window. The design of such trajectories is a problem of direct industrial relevance, and forms the central subject of this work.

1.1 Research Gap

Existing literature on trajectory design for wafer steppers evaluates candidate profiles exclusively through positioning metrics: tracking error, settling time, and the filtered exposure indices formalized in Chapter 3. What those metrics do not capture is the *morphology* of the failures into which these residual errors translate to on the wafer. Yield engineering, meanwhile, has long exploited that same morphology in the opposite direction: the pattern of the failing dies across the wafer — its *fingerprint* — is routinely used to diagnose which process step caused an observed defect[10], and industrial datasets such as WM-811K[16] catalogue hundreds of thousands of labelled examples of these patterns. No published methodology runs this relationship forward: from a given set of high-order kinematic bounds, predict the defect morphology they will produce on the wafer, and use that prediction as a cost function to optimize the trajectory. The absence of this link means that more aggressive acceleration profiles cannot be explored in an informed way, simultaneously limiting the achievable throughput and the wafer yield.

1.2 Hypothesis

The central hypothesis of this work is:

> *The kinematic bounds ($v_{\max}$, $a_{\max}$, $j_{\max}$, $s_{\max}$) of a fourth-order reference trajectory can be optimized, subject to a throughput constraint, using as cost function the probability distribution over defect-morphology classes estimated by a convolutional neural network trained on industrial wafer fingerprints.*

For the hypothesis to be testable, we decompose into three falsifiable sub-claims tied to a measurable criterion:

- **H1 (Sensitivity)** The exposure-window synchronization error of the simulated machine responds monotonically and reproducibly to each active kinematic bound.
- **H2 (Expressiveness)** The synthetic wafer defect maps produced from the simulation can be meaningfully separated by the CNN such that only the scanner-plausible classes activate.
- **H3 (Optimality)** A population-based search over the four kinematic bounds, driven by the CNN cost, converges to profiles that overcome third-order and regular fourth-order baselines on the yield–throughput plane.

H1 is formulated because if sweeping a bound leaves exposure indices unchanged (or changes them erratically) it provides no gradient to effectively guide an optimizer. Similarly, **H2** is driven by the need to use the CNN probability distribution as a criterion to isolate the phenomenological causes behind the resulting error type on the wafer, as well as ensuring the simulation captures real scanner physics. Furthermore, **H3** establishes that a yield-optimal trajectory profile can be iteratively approximated to push past conventional baseline limitations. Chapter 5 evaluates these three claims.

1.3 Objectives

The general objective is to develop a computational framework comprising a simulation of a lithography scanner and a convolutional neural network as a defect-evaluation model that can provide us with a yield–throughput plane over which a population-based optimizer can search for fourth-order trajectories that maximize estimated yield without penalizing throughput. This decomposes into six specific objectives:

O1 Implement the multi-body dynamic model of[2] as a simulation restricted to the x–y plane, capable of following a parameterized trajectory profile.

O2 Describe an analytical model to generate fourth-order trajectories from kinematic bounds ($v_{\max}$, $a_{\max}$, $j_{\max}$, $s_{\max}$) and segment times.

O3 Train and validate a convolutional neural network to classify the nine defect-morphology classes on WM-811K[16].

O4 Design a mapping from the dynamic error on a simulated traversal into a synthetic wafer defect map comparable with the CNN inputs.

O5 Implement a population-based optimizer coupled to the simulation using the CNN's probability distribution as a dynamic cost function to optimize the kinematic parameters of a reference trajectory.

O6 Evaluate the framework through comparative scenarios quantifying the throughput–yield trade-off against third-order and regular fourth-order baselines.

Objectives O1–O5 are implemented and validated in this document (Chapters 2–5), including the exact fifteen-segment snap-limited generator together with its second- and third-order counterparts (Section 4.3). O6 is addressed by the optimization scenarios E3 and E4 and by the profile-order comparison of Experiment 7.

1.4 Outline of the Thesis

Chapter 2 develops the formulation of the multi-body model of the machine in state-space. Chapter 3 translates the industrial drivers into quantitative control objectives. Chapter 4 describes the simulation environment, the trajectory generator, the stage controller, the CNN pipeline. Chapter 5 reports the experimentals performed to validate the simulated machine, the evaluation of the classifier and the trajectory search. The conclusion summarizes findings, limitations, and further comparative scenarios.

Chapter 2

Model

2.1 Introduction

As we established before, the accurate relative positioning between the reticle and the wafer is crucial for *step-and-scan* machines to produce functional ICs. To analyze these positioning requirements in a multi-body setting, we leverage existing literature[2] that proposes modeling the lithography scanner as a combination of three open-loop serial chains: the reticle chain ($j = 1$, blue), the optics chain ($j = 2$, purple), and the wafer chain ($j = 3$, red); the latter being the crux of the problem studied here.

The base frame serves as the common structural root for all three chains. In the reticle chain, a dual-stage architecture is employed to satisfy the competing requirements for travel range and precision. Intially, the long-stroke motor (LS) —of the planar moving-coil type[14]— consisting of four coil actuators arranged in a platform shape facilitates large-range travel (of at least 300mm) with moderate micrometer accuracy in 6-DOF. However, being directly mounted to the base frame, it is inherently susceptible to vibrations propagated from the rest of the machine.

To achieve nanometric tracking accuracy, a short-stroke motor (SS) is appended to the long-stroke carrier. Since travel in this stage is limited up to 1mm, it is realized with Lorentz actuators which provide contact-free, high-bandwidth positioning. Given that the Lorentz force depends entirely on the current within the actuator and is independent of the stator's motion, it effectively decouples the end-effector from the base frame disturbances through magnetic suspension[3]. To minimize linearity errors and optimize force transfer, the long-stroke stage carrying the stator must maintain the optimal clearance to the magnets of the rest of the short-stroke actuator. This arrangement is replicated for the wafer chain.

The optics chain is attached to the base frame through a metrology frame which is in turn supported by pneumatic vibration isolators[3]. While the optics box includes adaptive optics with multiple elements, it is reduced in this framework to a single fixed lens. This allows the model to focus on machine compliance and structural vibration transfer rather than involving active optical control[2].

The physical connections between these stages, which constitute the primary transmission path for vibration, are represented by lumped-parameter spring-damper systems, according to the Kelvin-Voigt model. As its proponents state, this low-fidelity modeling choice is sufficient to study the dynamics between the multiple bodies during *step-and-scan* motion[2].

Figure 2.1 illustrates the structural topology and hierarchical arrangement within each chain. Horizontal Kelvin–Voigt elements $(K_{i,j}, C_{i,j})$ couple each stage laterally to the fixed structural reference depicted as a solid line in the color of the corresponding chain; in particular, elements (K_1, C_1) provide passive vibration isolation between $\mathcal{F}_0 \rightarrow \mathcal{F}_1$ through the solid green line that extends the base frame $(i = 1)$ into the coupling to the first body $(i = 2)$ in any j chain. The in-plane deviations $f_{i,j}$ represented by the horizontal blue double-headed arrows measure the relative displacement between each body. The dashed vertical centerline represents the nominal reference for $f^0_{i,j}$, when all relative coordinates are zero.

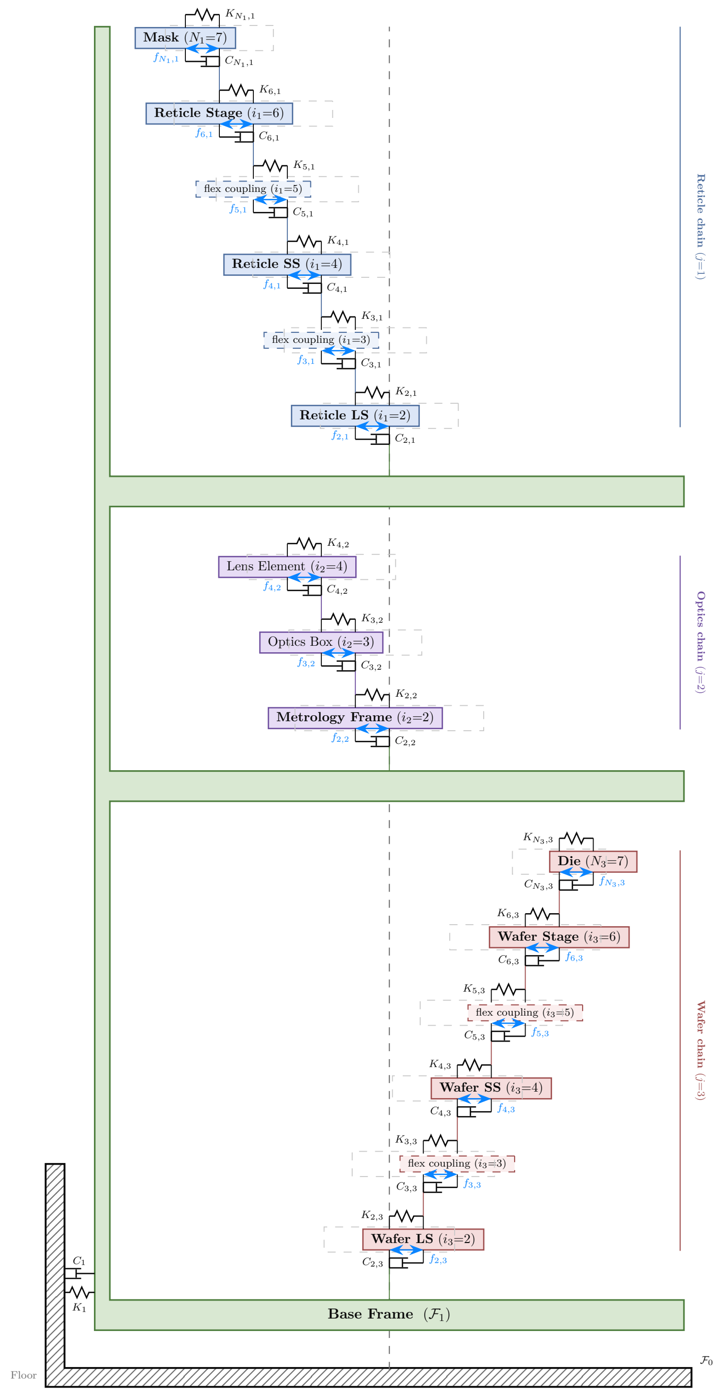


Figure 2.1: Lumped mass-spring-damper system of the three chains vertically stacked on the common base frame

2.2 Dynamics

The full 6-DOF model is here restricted to in-plane motions[1;4], since rotations about the x and y axis, as well as translation on z are used mainly for focusing. The generalized coordinate used to describe the configuration of the ith stage within the jth chain is

$$q_{i_j} = \begin{bmatrix} f_{i_j} \\ \theta_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3\times 1} \tag{2.2.1}$$

where $f_{i_j} = [f_{x,i_j},\ f_{y,i_j}]^T \in \mathbb{R}^2$ collects the in-plane translational displacements of stage i relative to its parent body in chain j, and $\theta_{z_{i_j}} \in \mathbb{R}$ is the corresponding rotation about the out-of-plane z-axis. The joints between bodies within a chain are modeled using a spring-damper system[2]. The associated generalized mass-inertia matrix for each body is

$$M_{i_j} = \begin{bmatrix} m_{i_j}\,\mathbf{I}_2 & 0 \\ 0 & I_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3\times 3} \tag{2.2.2}$$

and the generalized stiffness and damping matrices K_{i_j}, $C_{i_j} \in \mathbb{R}^{3\times 3}$ are block-diagonal, with a 2×2 translational block and a scalar rotational entry.

The absolute position $f^0_{i_j}$ of body i is expressed as the recursive sum of its relative displacement from its parent body along the chain from the floor frame $\mathcal{F}_0$. Thus, for bodies with $i \geq 2$ on any chain $j \in \{1, 2, 3\}$:

$$f^0_{i_j} = f^0_{(i-1)_j} + f_{i_j} \tag{2.2.3}$$

Since the rotation about the z-axis is a scalar, the angular position recurrence is simply additive:

$$\theta^0_{z_{i_j}} = \theta^0_{z_{(i-1)_j}} + \theta_{z_{i_j}} \tag{2.2.4}$$

From this, the translational velocity expression appears as:

$$\dot{f}^0_{i_j} = \dot{f}^0_{(i-1)_j} + \dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j} \tag{2.2.5}$$

Where $\omega_{(i-1)_j}$ is the angular velocity[5] of $\mathcal{F}_{(i-1)_j}$ with respect to the floor, $\mathcal{F}_0$.

Taking the second derivative gives:

$$
\begin{aligned}
\ddot{f}^0_{i_j} &= \ddot{f}^0_{(i-1)_j} + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \dot{\omega_{(i-1)_j}} \times f_{i_j} + \omega_{(i-1)_j} \times \left[\dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j}\right] \\
&= \ddot{f}^0_{(i-1)_j} + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \dot{\omega_{(i-1)_j}} \times f_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \omega_{(i-1)_j} \times \left[\omega_{(i-1)_j} \times f_{i_j}\right] \quad (2.2.6) \\
&= \ddot{f}^0_{(i-1)_j} + \ddot{f}_{i_j} + 2\left[\omega_{(i-1)_j} \times \dot{f}_{i_j}\right] + \dot{\omega_{(i-1)_j}} \times f_{i_j} + \omega_{(i-1)_j} \times \left[\omega_{(i-1)_j} \times f_{i_j}\right]
\end{aligned}
$$

Since we are concerned with planar movements in this work, we restrict the angular velocity of each body purely about the z-axis as $\omega = \dot{\theta}_z\, \hat{z}$, so that:

$$\omega_{(i-1)_j} = \dot{\theta}^0_{z_{(i-1)_j}}\, \hat{z}$$

We substitute the cross-products as matrix-vector products using the skew-symmetric matrix. In the 6-DOF model[2], it appears as:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Clearly, here we can set $\omega_x = 0$ and $\omega_y = 0$, thus $\omega = [0,\, 0,\, \dot{\theta}_z]^T$ and the matrix becomes:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & 0 \\ \omega_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Considering our 2D displacement vector, we only take the upper-left 2×2 sub-matrix, and factoring out the scalar $\dot{\theta}^0_{z_{(i-1)_j}}$:

$$[\omega_{(i-1)_j}] = \dot{\theta}^0_{z_{(i-1)_j}} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{2.2.7}$$

such that $\omega_{(i-1)_j} \times f_{i_j} = [\omega_{(i-1)_j}]\, f_{i_j}$. Applying this matrix twice gives the centripetal term:

$$[\omega_{(i-1)_j}]^2 = -\left(\dot{\theta}^0_{z_{(i-1)_j}}\right)^2 \mathbf{I}_2 \tag{2.2.8}$$

so that $\omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] = [\omega_{(i-1)_j}]^2\, f_{i_j}$. Substituting and expanding (2.2.3) recursively from the floor frame, the translational kinematics become:

$$f^0_{i_j} = f^0_1 + \sum_{k=2}^{i} f_{k_j} \tag{2.2.9}$$

$$\dot{f}^0_{i_j} = \dot{f}^0_1 + \sum_{k=2}^{i} \dot{f}_{k_j} + \sum_{k=2}^{i} [\omega_{(k-1)_j}]\, f_{k_j} \tag{2.2.10}$$

$$\ddot{f}^0_{i_j} = \ddot{f}^0_1 + \sum_{k=2}^{i} \ddot{f}_{k_j} + \sum_{k=2}^{i} \left[2\,[\omega_{(k-1)_j}]\,\dot{f}_{k_j} + [\dot{\omega}_{(k-1)_j}]\, f_{k_j} + [\omega_{(k-1)_j}]^2\, f_{k_j}\right] \tag{2.2.11}$$

Since the rotation $\theta_{z_{i_j}}$ is a scalar, no Coriolis or centripetal cross-terms arise in the angular kinematics. Expanding (2.2.4) recursively from the floor frame gives the rotational analogues:

$$\theta^0_{z_{i_j}} = \theta^0_{z_1} + \sum_{k=2}^{i} \theta_{z_{k_j}} \tag{2.2.12}$$

$$\dot{\theta}^0_{z_{i_j}} = \dot{\theta}^0_{z_1} + \sum_{k=2}^{i} \dot{\theta}_{z_{k_j}} \tag{2.2.13}$$

$$\ddot{\theta}^0_{z_{i_j}} = \ddot{\theta}^0_{z_1} + \sum_{k=2}^{i} \ddot{\theta}_{z_{k_j}} \tag{2.2.14}$$

Thus the equation of the angular motion of the ith body in the jth chain w.r.t $\mathcal{F}_0$ is:

$${}^0I_j\,\dot{\omega}_j + ([\omega_j])\,{}^0I_j\,\omega_j = m^0_j \tag{2.2.15}$$

Under the planar restriction $\omega_x = \omega_y = 0$ the gyroscopic term vanishes like:

$$\left([\omega_{i_j}]\,{}^0I_{i_j}\right)\omega_{i_j} = \begin{bmatrix} 0 & -\dot{\theta}_z & 0 \\ \dot{\theta}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_z \end{bmatrix} = 0$$

Leaving only:

$$I_{z_{i_j}}\,\ddot{\theta}^0_{z_{i_j}} = m^0_{z_{i_j}} \tag{2.2.16}$$

where $I_{z_{i_j}}$ is the aggregated moment of inertia of all bodies from i to the end-effector N_j, referred to the z-axis of body i via Steiner's theorem, and $m^0_{z_{i_j}} \in \mathbb{R}$ is the net applied moment acting on body i in chain j about the z-axis.

Steiner's theorem and inertia aggregation. The parallel axis (Steiner's) theorem states that the moment of inertia of a rigid body about any axis parallel to one through its center of mass (CoG) satisfies

$$I_z = \tilde{I}_z + m\,\|f\|^2, \tag{2.2.17}$$

where $\tilde{I}_z$ is the body's own moment of inertia about the z-axis through its CoG, m is its mass, and $\|f\|^2 = f_x^2 + f_y^2$ is the squared in-plane distance between the two parallel axes. In a serial chain, the inertia of every downstream body must be shifted to the frame of the current body using (2.2.17). Following the characterization article[2] (Algorithm 1 therein), the aggregated inertia for the jth chain is computed recursively from the end-effector inward. In the planar restriction, the full 3×3 inertia matrix of the original algorithm reduces to a scalar z-axis inertia:

Algorithm 1 Planar inertia aggregation for the jth chain — planar reduction of Algorithm 1 in[2]

Require: CoG inertias $\tilde{I}_{z_{i_j}}$, masses m_{i_j}, relative positions f_{k_j} for $i = 2, \dots, N_j$

Ensure: Aggregated inertias ${}^0I_{z_{i_j}}$ for $i = 2, \dots, N_j$

1: **Initialize:** ${}^0I_{z_{N_j}} \leftarrow \tilde{I}_{z_{N_j}}$
2: **if** $2 \le i < N_j$ **then**
3:     **for** $k = N_j,\ \dots,\ i+1$ **do**
4:         $\tilde{m}_{(k-1)_j} \leftarrow \sum_{b=k}^{N_j} m_{b_j}$ ▷ accumulated downstream mass
5:         ${}^0I_{z_{(k-1)_j}} \leftarrow {}^0I_{z_{k_j}} + \tilde{I}_{z_{(k-1)_j}} + \tilde{m}_{(k-1)_j}\left(f_{x,k_j}^2 + f_{y,k_j}^2\right)$ ▷ Steiner shift from frame k to frame $k-1$
6:     **end for**
7: **end if**

Here $\tilde{m}_{(k-1)_j} = \sum_{b=k}^{N_j} m_{b_j}$ is the total mass of all downstream bodies (from k to the end-effector), $\tilde{I}_{z_{(k-1)_j}}$ is the local inertia of body $k-1$ about its own CoG, and $\tilde{m}_{(k-1)_j}(f_{x,k_j}^2 + f_{y,k_j}^2)$ is the Steiner correction that shifts the accumulated inertia from body k's CoG to body $(k-1)$'s CoG. By the time the algorithm terminates, ${}^0I_{z_{2_j}}$ contains the inertia of the entire chain j

referred to the CoG of its first body.

The aggregated inertia of the base frame (body 1) applies the same shift one more time, from body 2's CoG to the base frame:

$$
{}^{0}I_{z_1} = \tilde{I}_{z_1} + \sum_{j=1}^{3}\Big({}^{0}I_{z_{2_j}} + \tilde{m}_{1_j}\big(f_{x,2_j}^2 + f_{y,2_j}^2\big)\Big), \tag{2.2.18}
$$

where $\tilde{m}_{1_j} = \sum_{b=2}^{N_j} m_{b_j}$ is the total mass of chain j. This is the quantity denoted I_{z_0} in (2.2.22).

Expanding $\ddot{\theta}^0_{z_{i_j}}$ using (2.2.14):

$$
I_{z_{i_j}}\ddot{\theta}^0_{z_1} + I_{z_{i_j}}\sum_{k=2}^{i}\ddot{\theta}_{z_{k_j}} = m^0_{z_{i_j}} \tag{2.2.19}
$$

2.2.1 For the base frame

To write the equations that govern the base frame, we start from Newton's equation $F = ma$. For every body ($i \geq 2$) within chain j, it connects to body $i-1$ through a spring above and body $i+1$ through the spring below:

$$
\begin{gathered}
m_{i_j}\ddot{f}^0_1 + m_{i_j}\sum_{k=2}^{i}\ddot{f}_{k_j} + m_{i_j}\sum_{k=2}^{i}\Big[2\left[\omega_{(k-1)_j}\right]\dot{f}_{k_j} + \left[\dot{\omega}_{(k-1)_j}\right]f_{k_j} + \left[\omega_{(k-1)_j}\right]^2 f_{k_j}\Big] \\
-K_i\left(f^0_{(i-1)_j} - f^0_{i_j}\right) - C_i\left(\dot{f}^0_{(i-1)_j} - \dot{f}^0_{i_j}\right) + K_{i+1}\left(f^0_{i_j} - f^0_{(i+1)_j}\right) + C_{i+1}\left(\dot{f}^0_{i_j} - \dot{f}^0_{(i+1)_j}\right) = u^0_{i_j}
\end{gathered} \tag{2.2.20}
$$

Summing (2.2.20) over all bodies in a chain (taking the optics chain $j = 2$, bodies $i = 2, 3, 4$ as a representative example), the internal spring forces cancel by Newton's third law:

$$
\begin{aligned}
&- K_2(f^0_1 - f^0_2) - C_2(\dot{f}^0_1 - \dot{f}^0_2) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad 0 \\
&+K_3(f^0_2 - f^0_3) + C_3(\dot{f}^0_2 - \dot{f}^0_3) - K_3(f^0_2 - f^0_3) - C_3(\dot{f}^0_2 - \dot{f}^0_3) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad 0 \\
&+K_4(f^0_3 - f^0_4) + C_4(\dot{f}^0_3 - \dot{f}^0_4) - K_4(f^0_3 - f^0_4) - C_4(\dot{f}^0_3 - \dot{f}^0_4)
\end{aligned}
$$

Only the boundary term $-K_2(f_1^0 - f_2^0) - C_2(\dot{f}_1^0 - \dot{f}_2^0)$ survives, which corresponds to the connection between body $i = 2$ and the base frame. This extends to any N_j-body chain. Recalling (2.2.3), $f_1^0 - f_{2_j}^0 = -f_{2_j}$, and following[2] with ${}^0K_{i_j} = {}^0R_{i_j} K_{i_j} {}^0R_{i_j}^T$ and ${}^0C_{i_j} = {}^0R_{i_j} C_{i_j} {}^0R_{i_j}^T$, the total translational equation of the base frame (absorbing the balance masses and collecting all chain contributions) is:

$$m_0\, \ddot{f}_1^0 + {}^0\bar{C}_1\, \dot{f}_1 + {}^0\bar{K}_1\, f_1 - \sum_{j=1}^{3} \left\{ {}^0\hat{C}_{2_j}\, \dot{f}_{2_j} + {}^0\hat{K}_{2_j}\, f_{2_j} \right\} = u_1^0 \tag{2.2.21}$$

where $m_0 = m_1 + \sum_{j=1}^{3} \left(m_B^j + \sum_{i=2}^{N_j} m_{i_j} \right)$ is the total system mass, and the stiffness and damping matrices denote the 2×2 translational block. Following[2], ${}^0\bar{K}_1 = {}^0K_1$ and ${}^0\bar{C}_1 = {}^0C_1$, while the hatted chain couplings are ${}^0\hat{C}_{2_j} = {}^0C_{2_j}$ and ${}^0\hat{K}_{2_j} = {}^0K_{2_j} + {}^0C_{2_j}[\omega_1]$; the accents are defined in general in (2.2.29) below.

By the exact same cancellation argument applied to the moment balance about the z-axis — the moment exerted by the spring connecting body i to body $i-1$ is $-K_{i_j}\, \theta_{z_{i_j}}$, so internal moments cancel identically — the rotational equation of the base frame is:

$$I_{z_0}\, \ddot{\theta}_{z_1}^0 + {}^0\bar{C}_1\, \dot{\theta}_{z_1} + {}^0\bar{K}_1\, \theta_{z_1} - \sum_{j=1}^{3} \left\{ {}^0\hat{C}_{2_j}\, \dot{\theta}_{z_{2_j}} + {}^0\hat{K}_{2_j}\, \theta_{z_{2_j}} \right\} = \tau_1^0 \tag{2.2.22}$$

where $I_{z_0} = I_{z_1} + \sum_{j=1}^{3} \left(I_{z_B}^j + \sum_{i=2}^{N_j} I_{z_{i_j}} \right)$ is the total moment of inertia of the system about the z-axis, ${}^0\bar{C}_{i_j}$ and ${}^0\bar{K}_{i_j}$ (and their hatted counterparts, which coincide with them for the scalar rotational entries) are the scalar rotational damping and stiffness of the joint connecting body i to its predecessor in chain j, and τ_1^0 is the net applied torque on the base frame.

2.2.2 For each chain

Recalling (2.2.20) and (2.2.19), for the first body in the chain ($i = 2$), which connects to the base frame above and to body $i = 3$ below, the translational equation is:

$$\bar{m}_{2_j} \ddot{f}_{2_j}^0 + {}^0\bar{C}_{2_j}\, \dot{f}_{2_j} + {}^0\bar{K}_{2_j}\, f_{2_j} - \left\{ {}^0\hat{C}_{3_j}\, \dot{f}_{3_j} + {}^0\hat{K}_{3_j}\, f_{3_j} \right\} = u_{2_j}^0 \tag{2.2.23}$$

and the rotational equation is:

$$I_{z_{2_j}} \ddot{\theta}^0_{z_{2_j}} + {}^0\bar{C}_{2_j}\, \dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j}\, \theta_{z_{2_j}} - {}^0\hat{C}_{3_j}\, \dot{\theta}_{z_{3_j}} - {}^0\hat{K}_{3_j}\, \theta_{z_{3_j}} = \tau^0_{2_j} \tag{2.2.24}$$

For bodies $3 \le i < N_j$, the same structure holds with bodies $2, \ldots, i-1$ in between. The translational equation is:

$$\bar{m}_{i_j} \ddot{f}^0_{i_j} + {}^0\bar{C}_{i_j}\, \dot{f}_{i_j} + {}^0\bar{K}_{i_j}\, f_{i_j} - \left\{ {}^0\hat{C}_{(i+1)_j}\, \dot{f}_{(i+1)_j} + {}^0\hat{K}_{(i+1)_j}\, f_{(i+1)_j} \right\} = u^0_{i_j} \tag{2.2.25}$$

and the rotational equation is:

$$I_{z_{i_j}} \ddot{\theta}^0_{z_{i_j}} + {}^0\bar{C}_{i_j}\, \dot{\theta}_{z_{i_j}} + {}^0\bar{K}_{i_j}\, \theta_{z_{i_j}} - {}^0\hat{C}_{(i+1)_j}\, \dot{\theta}_{z_{(i+1)_j}} - {}^0\hat{K}_{(i+1)_j}\, \theta_{z_{(i+1)_j}} = \tau^0_{i_j} \tag{2.2.26}$$

Finally, for body $i = N_j$ (the end effector), there is no subsequent body in the chain. The translational and rotational equations are:

$$\bar{m}_{N_j j} \ddot{f}^0_{N_j j} + {}^0\bar{C}_{N_j j}\, \dot{f}_{N_j j} + {}^0\bar{K}_{N_j j}\, f_{N_j j} = 0 \tag{2.2.27}$$

$$I_{z_{N_j j}} \ddot{\theta}^0_{z_{N_j j}} + {}^0\bar{C}_{N_j j}\, \dot{\theta}_{z_{N_j j}} + {}^0\bar{K}_{N_j j}\, \theta_{z_{N_j j}} = 0 \tag{2.2.28}$$

2.2.3 Accented matrices and the nominal condition

Following Appendix A of[2], the accented matrices appearing in the equations above are not merely rotated joint matrices: they absorb the rotational cross-terms of (2.2.11) into effective damping and stiffness. With $[\omega]$ the planar skew form of (2.2.7) and $B_{i_j} = [\dot{\omega}_{i_j}] + [\omega_{i_j}]^2$:

$$\begin{aligned}
\bar{m}_{i_j} &= \sum_{k=i}^{N_j} m_{k_j} \\
{}^0\bar{C}_{i_j} &= {}^0C_{i_j} + 2m_{i_j}\, [\omega_{(i-1)_j}] \\
{}^0\bar{K}_{i_j} &= {}^0K_{i_j} + {}^0C_{i_j}\, [\omega_{(i-1)_j}] + m_{i_j}\, B_{(i-1)_j} \\
{}^0\hat{C}_{(i+1)_j} &= {}^0C_{(i+1)_j} \\
{}^0\hat{K}_{(i+1)_j} &= {}^0K_{(i+1)_j} + {}^0C_{(i+1)_j}\, [\omega_{i_j}]
\end{aligned} \tag{2.2.29}$$

The bar thus marks the matrices acting on the body's *own* joint coordinates, which collect the Coriolis ($2m[\omega]$), damper-convection ($C[\omega]$), and Euler–centripetal (mB) contributions, while the hat marks the coupling to the *downstream* joint coordinates, which inherits only the convective damper term: the damper reacts to the inertial velocity $\dot{f} + [\omega]f$ of (2.2.10), so part of its force is proportional to position and migrates into the effective stiffness. The scalar rotational entries admit no such cross-terms in the planar restriction, so about the z-axis the accents are inert and $\hat{(\cdot)} = \bar{(\cdot)} = (\cdot)$. For $3 \leq i < N_j$ the model of[2] additionally couples body i to every upstream joint through the two-index matrices ${}^0\hat{C}_{i_j,k} = 2m_{i_j}\,[\omega_{(k-1)_j}]$ and ${}^0\hat{K}_{i_j,k} = m_{i_j}\,B_{(k-1)_j}$ for $2 \leq k < i$, omitted above.

Every one of these corrections is proportional to $\dot{\theta}_z$ or $\ddot{\theta}_z$ of a parent body. At the nominal operating condition $\dot{\theta}_z = \ddot{\theta}_z = 0$ — the regulated step-and-scan regime studied throughout this work — they vanish identically: the accented matrices reduce to the frame-rotated ${}^0K_{i_j}$ and ${}^0C_{i_j}$, the inertial accelerations of (2.2.11) reduce to sums of relative accelerations, and the equations of motion coincide with Eqs. (2.6)–(2.8) of[2] and become linear time-invariant. Away from the nominal condition, the corrections re-enter as the parameter-varying part $A_1(\Omega, \dot{\Omega})$ of the decomposition given in[2]. Note that the scheduling variables Ω are themselves state variables — the $\dot{\theta}_z$ of each body — so the cross-terms are quadratic in the state (and the rotated matrices ${}^0K_{i_j}$, ${}^0C_{i_j}$ carry a further state dependence through ${}^0R_{i_j}(\theta_z)$): the machine model is, strictly, nonlinear, a *quasi*-LPV system whose frozen nominal dynamics are the LTI model above. Of the two realizations of the machine used in this work, the reduced two-body-per-chain model integrates this nominal LTI form directly, while the MuJoCo test bed assembles the full nonlinear multibody dynamics from the same lumped topology and therefore retains these couplings without approximation.

2.2.4 State-space formulation

Because each body possesses three planar degrees of freedom (f_x, f_y, θ_z) along with their corresponding time derivatives, it contributes exactly six states to the system. Following the framework outlined in Appendix B of[2], the complete machine model comprises 6 bodies in the reticle chain ($N_1 = 7$), 3 bodies in the optics chain ($N_2 = 4$), and 6 bodies in the wafer chain ($N_3 = 7$). Including the common base frame (body 1), the system features a total of $1 + (N_1 - 1) + (N_2 - 1) + (N_3 - 1) = 16$ bodies. Consequently, the global state vector x encompasses $6 \times 16 = 96$ states:

$$x = \Big[\underbrace{x_1, \ldots, x_6}_{\text{base frame}},\ \underbrace{x_7, \ldots, x_{6N_1}}_{\text{chain 1: bodies } 2 \to N_1},\ \underbrace{\cdots}_{\text{chain 2}},\ \underbrace{\cdots}_{\text{chain 3}}\Big]^T \tag{2.2.30}$$

The explicit mapping for each state is detailed in the table below. Here, p denotes the global sequential index of the i-th body in the j-th chain within the concatenated list of all bodies (starting with the base frame, followed sequentially by chains 1, 2, and 3). It is crucial to note that every positional entry within the state vector represents a *relative* joint displacement; the corresponding absolute inertial position $f^0_{i_j}$ can be reconstructed using the recursive summations from (2.2.9) to (2.2.12).

Solving (2.2.21) and (2.2.22) for the base-frame accelerations yields:

$$\ddot{f}^0_1 = -\frac{{}^0\bar{C}_1}{m_0}\,\dot{f}_1 - \frac{{}^0\bar{K}_1}{m_0}\,f_1 + \frac{1}{m_0}\sum_{j=1}^{3}\left\{{}^0\bar{C}_{2_j}\,\dot{f}_{2_j} + {}^0\bar{K}_{2_j}\,f_{2_j}\right\} + \frac{u^0_1}{m_0} \tag{2.2.31}$$

$$\ddot{\theta}^0_{z_1} = -\frac{{}^0\bar{C}_1}{I_{z_0}}\,\dot{\theta}_{z_1} - \frac{{}^0\bar{K}_1}{I_{z_0}}\,\theta_{z_1} + \frac{1}{I_{z_0}}\sum_{j=1}^{3}\left\{{}^0\bar{C}_{2_j}\,\dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j}\,\theta_{z_{2_j}}\right\} + \frac{\tau^0_1}{I_{z_0}} \tag{2.2.32}$$

Table 2.2.1: State-space mapping for the global concatenated system vector

State-space Variable	Expression	Meaning
x_1	f_{x_1}	base frame x-position
x_2	f_{y_1}	base frame y-position
x_3	θ_{z_1}	base frame rotation
x_4	$\dot{f}_{x_1}$	base frame x-velocity
x_5	$\dot{f}_{y_1}$	base frame y-velocity
x_6	$\dot{\theta}_{z_1}$	base frame angular velocity
x_7	$f_{x_{2_1}}$	chain 1, body 2, x-position
x_8	$f_{y_{2_1}}$	chain 1, body 2, y-position
x_9	$\theta_{z_{2_1}}$	chain 1, body 2, rotation
x_{10}	$\dot{f}_{x_{2_1}}$	chain 1, body 2, x-velocity
x_{11}	$\dot{f}_{y_{2_1}}$	chain 1, body 2, y-velocity
x_{12}	$\dot{\theta}_{z_{2_1}}$	chain 1, body 2, angular velocity
$\vdots$	$\vdots$	$\vdots$
x_{6p-5}	$f_{x_{i_j}}$	chain j, body i, x-position
x_{6p-4}	$f_{y_{i_j}}$	chain j, body i, y-position
x_{6p-3}	$\theta_{z_{i_j}}$	chain j, body i, rotation
x_{6p-2}	$\dot{f}_{x_{i_j}}$	chain j, body i, x-velocity
x_{6p-1}	$\dot{f}_{y_{i_j}}$	chain j, body i, y-velocity
x_{6p}	$\dot{\theta}_{z_{i_j}}$	chain j, body i, angular velocity
$\vdots$	$\vdots$	$\vdots$

Expressed in the standard matrix form $\dot{x} = Ax + Bu$, the first six rows governing the base-frame dynamics are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \cdots \\ -\frac{K_{1,xx}}{m_0} & -\frac{K_{1,xy}}{m_0} & 0 & -\frac{C_{1,xx}}{m_0} & -\frac{C_{1,xy}}{m_0} & 0 & \frac{K_{2_1,xx}}{m_0} & \frac{K_{2_1,xy}}{m_0} & 0 & \frac{C_{2_1,xx}}{m_0} & \cdots \\ -\frac{K_{1,yx}}{m_0} & -\frac{K_{1,yy}}{m_0} & 0 & -\frac{C_{1,yx}}{m_0} & -\frac{C_{1,yy}}{m_0} & 0 & \frac{K_{2_1,yx}}{m_0} & \frac{K_{2_1,yy}}{m_0} & 0 & \frac{C_{2_1,yx}}{m_0} & \cdots \\ 0 & 0 & -\frac{K_{1,zz}}{I_{z_0}} & 0 & 0 & -\frac{C_{1,zz}}{I_{z_0}} & 0 & 0 & \frac{K_{2_1,zz}}{I_{z_0}} & 0 & \cdots \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_0} \\ 0 \\ 0 \end{bmatrix} \tag{2.2.33}$$

General State-Space Form for the Chain Bodies

To systematically construct the global state-space matrices, we must reframe the equations of motion for the chain bodies in terms of the state vector variables. We map the physical planar coordinates of the p-th sequential body to the local state vectors: translational position $\mathbf{x}_{2p-1} = [x_{6p-5}, x_{6p-4}]^T$, translational velocity $\dot{\mathbf{x}}_{2p-1} = [x_{6p-2}, x_{6p-1}]^T$, rotational position $\mathbf{x}_{2p} = x_{6p-3}$, and rotational velocity $\dot{\mathbf{x}}_{2p} = x_{6p}$.

The state derivatives are dictated by the relative accelerations between adjacent bodies. By applying the kinematic identity $\ddot{f}_{i_j} = \ddot{f}^0_{i_j} - \ddot{f}^0_{(i-1)_j}$ from (2.2.11), we determine the interactive forces acting on each body within its parent's reference frame. Let p represent the global index of body i in chain j, while $p-1$ and $p+1$ denote the indices of its immediate upstream and downstream neighbors respectively (with the base frame positioned at $p = 1$).

For the **first body in each chain** ($i = 2$, index p, upstream index 1), the translational state derivatives ($\ddot{\mathbf{x}}_{2p-1} = \ddot{f}_{2_j}$) and rotational state derivatives ($\ddot{\mathbf{x}}_{2p} = \ddot{\theta}_{z_{2_j}}$) take the form:

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{2_j}}\Big[u^0_{2_j} - {}^0\bar{C}_{2_j}\dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{2_j}\mathbf{x}_{2p-1} + {}^0\hat{C}_{3_j}\dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{3_j}\mathbf{x}_{2p+1}\Big] \\
&\quad - \frac{1}{m_0}\Big[u^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_1 - {}^0\bar{K}_1\mathbf{x}_1 + \sum_{k=1}^{3}\Big({}^0\hat{C}_{2_k}\dot{\mathbf{x}}_{2p_{2_k}-1} + {}^0\hat{K}_{2_k}\mathbf{x}_{2p_{2_k}-1}\Big)\Big]
\end{aligned} \tag{2.2.34}$$

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{2_j}}}\Big[\tau^0_{2_j} - {}^0\bar{c}_{\theta,2_j}\dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,2_j}\mathbf{x}_{2p} + {}^0\hat{c}_{\theta,3_j}\dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,3_j}\mathbf{x}_{2p+2}\Big] \\
&\quad - \frac{1}{I_{z_0}}\Big[\tau^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_2 - {}^0\bar{K}_1\mathbf{x}_2 + \sum_{k=1}^{3}\Big({}^0\hat{c}_{\theta,2_k}\dot{\mathbf{x}}_{2p_{2_k}} + {}^0\hat{k}_{\theta,2_k}\mathbf{x}_{2p_{2_k}}\Big)\Big]
\end{aligned} \tag{2.2.35}$$

For a **general intermediate body** ($2 < i < N_j$, index p), the dynamics are shaped by upstream, local, and downstream interactions:

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{i_j}}\Big[u^0_{i_j} - {}^0\bar{C}_{i_j}\dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{i_j}\mathbf{x}_{2p-1} + {}^0\hat{C}_{(i+1)_j}\dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{(i+1)_j}\mathbf{x}_{2p+1}\Big] \\
&\quad - \frac{1}{\bar{m}_{(i-1)_j}}\Big[u^0_{(i-1)_j} - {}^0\bar{C}_{(i-1)_j}\dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(i-1)_j}\mathbf{x}_{2p-3} + {}^0\hat{C}_{i_j}\dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{i_j}\mathbf{x}_{2p-1}\Big]
\end{aligned} \tag{2.2.36}$$

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{i_j}}}\Big[\tau^0_{i_j} - {}^0\bar{c}_{\theta,i_j}\dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,i_j}\mathbf{x}_{2p} + {}^0\hat{c}_{\theta,(i+1)_j}\dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,(i+1)_j}\mathbf{x}_{2p+2}\Big] \\
&\quad - \frac{1}{I_{z_{(i-1)_j}}}\Big[\tau^0_{(i-1)_j} - {}^0\bar{c}_{\theta,(i-1)_j}\dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(i-1)_j}\mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,i_j}\dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,i_j}\mathbf{x}_{2p}\Big]
\end{aligned} \tag{2.2.37}$$

Finally, for the **end-effector body** ($i = N_j$, index p), the downstream force terms naturally vanish as there are no subsequent bodies:

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{N_{j_j}}}\Big[u^0_{N_{j_j}} - {}^0\bar{C}_{N_{j_j}}\dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{N_{j_j}}\mathbf{x}_{2p-1}\Big] \\
&\quad - \frac{1}{\bar{m}_{(N_j-1)_j}}\Big[u^0_{(N_j-1)_j} - {}^0\bar{C}_{(N_j-1)_j}\dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(N_j-1)_j}\mathbf{x}_{2p-3} + {}^0\hat{C}_{N_{j_j}}\dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{N_{j_j}}\mathbf{x}_{2p-1}\Big]
\end{aligned} \tag{2.2.38}$$

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{N_{j_j}}}}\Big[\tau^0_{N_{j_j}} - {}^0\bar{c}_{\theta,N_{j_j}}\dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,N_{j_j}}\mathbf{x}_{2p}\Big] \\
&\quad - \frac{1}{I_{z_{(N_j-1)_j}}}\Big[\tau^0_{(N_j-1)_j} - {}^0\bar{c}_{\theta,(N_j-1)_j}\dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(N_j-1)_j}\mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,N_{j_j}}\dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,N_{j_j}}\mathbf{x}_{2p}\Big]
\end{aligned} \tag{2.2.39}$$

Chapter 3

Control Objectives

3.1 Introduction

In Chapter 1 we identified overlay and throughput as the two primary drivers of scanner design, followed in Chapter 2 by the modeling of its multi-body dynamics. This chapter focuses on mapping those industrial requirements into quantitative objectives that any candidate reference trajectory must satisfy. Here we explore in detail how the synchronization error arises between the wafer and the reticle stages, examine the filtered metrics through which the industry evaluates it, and establish the formal statement of the optimization problem.

3.2 Synchronization Error

During the scanning exposure, the projection optics image the reticle onto the wafer with a demagnification factor of $\alpha = 0.25$. Thus, the image of the reticle pattern is stationary on the wafer surface only if the wafer follows the reticle motion scaled by α at every instant. The degradation of the printed pattern is therefore not due to the individual tracking error of either stage, but their scaled difference. Let $r_w(t)$ and $r_r(t)$ denote the reference positions of the wafer and reticle stages respectively along the scanning direction, and $y_w(t)$, $y_r(t)$ their measured positions. The *synchronization error* is defined as[3]

$$e_{syn}(t) = \big(y_w(t) - r_w(t)\big) - \alpha\big(y_r(t) - r_r(t)\big). \tag{3.2.1}$$

Naturally, a common-mode mode disturbance that displaces both stages consistently with the projection ratio leaves e_{syn} unaffected. Conversely, any differential motion between the chains — exacerbated by the lightly damped structural resonance — is transferred directly to the printed image.

3.3 Exposure Metrics: MA and MSD

For an area of photoresist to become exposed, the illumination dose must overcome a threshold. Despite direct exposure during the scanning motion, the photoresist on the wafer does not respond to the instantaneous value of e_{syn} but to its aggregate effect over the time a given point stays below the light source. Therefore, industry practice evaluates the synchronization error through two moving-window filters matched to the exposure time t_{exp} of one point on the wafer[3]:

- First, the **moving average**

$$\mathrm{MA}(t_k) = \frac{1}{N}\sum_{n=0}^{N-1} e_{syn}(t_{k-n}) \tag{3.3.1}$$

  measures the mean displacement of the image and manifests as *overlay* error or the misalignment of the printed pattern with respect to the previous layer.

- Second, the **moving standard deviation**

$$\mathrm{MSD}(t_k) = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1}\Big(e_{syn}(t_{k-n}) - \mathrm{MA}(t_k)\Big)^2} \tag{3.3.2}$$

  measures the smearing of the aerial image around that mean position and manifests as *fading*, a loss of contrast that distorts the printed feature.

For the 38 nm half-pitch process, the specifications adopted[2] are

$$\text{MA} \in [-1.25,\ +2]\ \text{nm}, \qquad \text{MSD} \leq 7\ \text{nm}, \tag{3.3.3}$$

evaluated at every instant during the exposure window[3]. These bounds must be met while the stage sustains the required scanning velocity, operating within a settling time budget of under 10 ms to preserve wafer throughput

3.4 Problem Statement

The reference trajectories studied here are fourth-order profiles, parameterized by the kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ on velocity, acceleration, jerk, and snap. These four scalars determine both the duration of the motion — and hence the throughput in wafers per hour — and the spectral content of the excitation injected into the machine structure, and hence the synchronization error during exposure.

Conventional trajectory design evaluates candidate parameters against servo-mechanical metrics: settling time, peak error, or the MA/MSD bounds of (3.3.3). Such metrics declare a die good or bad, but say nothing about the *morphology* of the failures that accumulate across the wafer. Wafer fabs routinely exploit the converse relationship: the spatial pattern of failing dies — the wafer *fingerprint* — is used to diagnose which process step caused an observed defect, an inverse problem well established in industry[10]. The forward direction, however, remains open: no published methodology predicts, from a given set of trajectory parameters, the defect pattern they will produce, nor uses that prediction as the cost function during trajectory optimization. This thesis addresses precisely that forward direction.

To quantify the yield impact of a candidate trajectory, the defect map produced under that trajectory is classified by a convolutional neural network trained on industrial wafer maps, which estimates a probability P_c for each defect morphology class c. The cost assigned to the trajectory is the expected yield loss

$$\mathcal{J} = \sum_{c \in \mathcal{C}} w_c P_c, \tag{3.4.1}$$

where w_c is the economic weight of class c[8;9;7] and $\mathcal{C}$ is restricted to the classes plausibly caused by scanner motion — *none*, *scratch*, *edge-local*, and *random*. The remaining classes of the taxonomy (center, donut, edge-ring, near-full) originate in etching, cleaning, and deposition steps, and their activation by a simulated map would indicate a modeling error rather than a trajectory effect; this restriction is used as a validation criterion for the trajectory-to-defect mapping in Chapter 5.

The optimization problem is then stated as

$$\min_{(v_{\max}, a_{\max}, j_{\max}, s_{\max})} \mathcal{J} \qquad \text{subject to} \qquad T_{wafer} \leq T_{budget}, \tag{3.4.2}$$

where T_{wafer} is the total motion time per wafer implied by the trajectory parameters and T_{budget} is the cycle-time budget corresponding to the target throughput. The hypothesis of this work is that (3.4.2) can be solved with a data-driven cost of the form (3.4.1), producing trajectory parameters that trade throughput against expected yield rather than against servo metrics alone.

Chapter 4

Methodology

4.1 Introduction

Trajectory generation is a well studied problem. We understand that trajectories that have smooth higher order derivatives result in less excitation of poorly damped modes in the structure. To properly test the trajectories we will be generating against the yield they will generate on the machine, then we need a test bed on which they can be performed on-line before actually reproducing them in the machine.

4.2 Simulation

We implemented a simulation of the model we recovered from the state-space characterization article[2] using the MuJoCo[15] physics engine for Python. The simulation serves as a virtual test bed where the trajectories proposed can be validated under the dynamic model we developed in chapter 2.

The MuJoCo Python bindings allows us to define the context of our simulation from an XML string by calling the `mujoco.MjModel.from_xml_string(MODEL_XML)` method. We extend from

MuJoCo's `<worldbody>` tag on which we place a root `<body>` tag, representing our *base frame*; this allows us to organize the machine in the simulation using the three chain hierarchical tree. Table 4.2.1 maps the specific names used in the MuJoCo XML to their counterparts in the structural diagram of Figure 2.1.

Table 4.2.1: Mapping of MuJoCo XML bodies to the model components

`<body>` `name` property	Model body
`base_frame`	Base Frame ($\mathcal{F}_1$)
`w_ls_act`	Wafer LS ($i_3 = 2$)
`w_ls_flex`	flex coupling ($i_3 = 3$)
`w_ss_act`	Wafer SS ($i_3 = 4$)
`w_ss_flex`	flex coupling ($i_3 = 5$)
`w_st_act`	Wafer Stage ($i_3 = 6$)
`wafer`	Die ($N_3 = 7$)
`r_ls_act`	Reticle LS ($i_1 = 2$)
`r_ls_flex`	flex coupling ($i_1 = 3$)
`r_ss_act`	Reticle SS ($i_1 = 4$)
`r_ss_flex`	flex coupling ($i_1 = 5$)
`r_st_act`	Reticle Stage ($i_1 = 6$)
`mask`	Mask ($N_1 = 7$)
`metro_frame`	Metrology Frame ($i_2 = 2$)
`optics_box`	Optics Box ($i_2 = 3$)
`lens`	Lens Element ($i_2 = 4$)

The simulation environment is defined through several specialized XML attributes that handle different aspects of the physical world. Table 4.2.2 summarizes the function of each tag within the context of our lithography scanner model.

Table 4.2.2: MuJoCo XML tag functions in the scanner simulation

Tag	Function
`<worldbody>`	Defines the inertial root, the floor plane ($\mathcal{F}_0$).
`<body>`	Creates a rigid body with its own local coordinate system.
`<inertial>`	Assigns mass m_{i_j} and the aggregated inertia tensor I_{i_j}.
`<joint>`	Implements the degrees of freedom f_{i_j} and $\theta_{z_{i_j}}$.
`<geom>`	Defines the material (granite, silicon, glass) and collision shapes.
`<actuator>`	Models the Lorentz and moving-coil motors via servo controllers.

The mechanical properties of the couplings, represented by the Kelvin-Voigt elements, are implemented via the `stiffness` and `damping` attributes of the `<joint>` tags. For actuated stages, where the physical stiffness is replaced by the controller's active response, the simulation uses high `kp` values. Tables 4.2.3, 4.2.4, and 4.2.5 illustrate the distribution of these variables for the three chains.

Table 4.2.3: Wafer chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor → Base Frame	`K1`	`C1`
Base → LS Stage	`KP_ACT`	`KV_ACT_TRANS`
LS Stage → LS Flex	`K_LS_F`	`C_LS_F`
SS Stage → SS Flex	`K_SS_F`	`C_SS_F`
Fine Stage → Flex	`K_ST_F`	`C_ST_F`
Stage → Die	`K8_W`	`C8_W`

As we mentioned, the simulation topology mirrors the multi-body hierarchy described in Chapter 2. The common base frame ($\mathcal{F}_1$) is anchored to the world frame ($\mathcal{F}_0$) via a 3-DOF joint (two prismatic, one revolute) representing the isolation system with stiffness K_1 and damping

Table 4.2.4: Reticle chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor → Base Frame	`K1`	`C1`
Base → LS Stage	`KP_ACT`	`KV_ACT_TRANS`
LS Stage → LS Flex	`K_LS_F`	`C_LS_F`
SS Stage → SS Flex	`K_SS_F`	`C_SS_F`
Fine Stage → Flex	`K_ST_F`	`C_ST_F`
Stage → Mask	`K8_R`	`C8_R`

Table 4.2.5: Optics chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Base Frame → Metro Frame	`K2_O`	`C2_O`
Metro Frame → Optics Box	`K3_O`	`C3_O`
Optics Box → Lens Element	`K4_O`	`C4_O`

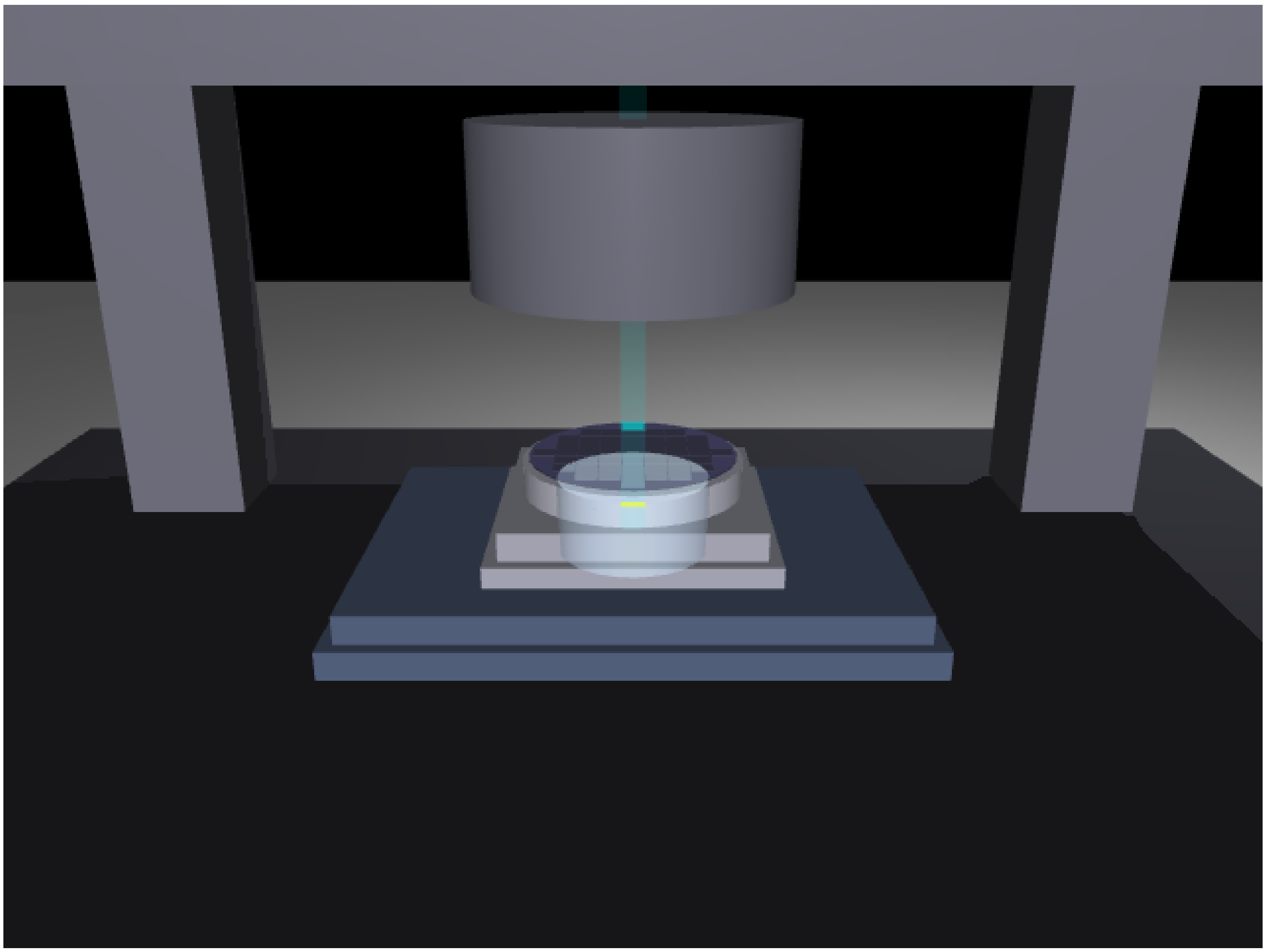


Figure 4.1: Screenshot of the MuJoCo simulation in the midst of the scanning motion, showing the wafer and the optics chain under the metrology frame

C_1 as seen in Eq. (2.2.21) and (2.2.22). From this base, three serial chains are constructed using nested body definitions in MuJoCo's XML format, where each child body's joint coordinate corresponds directly to the relative displacement f_{i_j} and rotation $\theta_{z_{i_j}}$ defined in our kinematics model.

The passive Kelvin-Voigt couplings $(K_{i,j}, C_{i,j})$ are implemented as MuJoCo joints with explicit stiffness and damping parameters. For the actuated stages (Long-Stroke, Short-Stroke, and Fine stages), we use MuJoCo's position actuators with proportional gain $K_P = 10^7\,\mathrm{N/m}$, chosen to approximate the stiffness of a rigid servo motor while maintaining numerical stability with the implicit integrator at a timestep of $dt = 10^{-4}\,\mathrm{s}$.

Two properties of the reference machine model[2] require care in this translation, because MuJoCo's native primitives do not reproduce them by default. First, the guideways of the actuated stages are *frictionless*: the stages ride on air bearings, and the reference model explicitly discards all sliding and rolling resistance. The servo velocity gain K_V must therefore act on the *tracking-error* velocity, not on the joint velocity itself. For the long-stroke joints, whose reference moves at the scan speed, this is implemented with velocity actuators commanded with the reference velocity, contributing $K_V(\dot{r} - \dot{y})$; implementing K_V as joint damping would instead exert a drag force $K_V \cdot v$ against the position servo during cruise — a synthetic servo lag of $K_V/K_P = 10\,\mathrm{mm}$ per m/s with no physical counterpart in an air-bearing stage. Joints that regulate a constant reference (the short-stroke and fine stages, and all rotations) keep plain joint damping, for which the two forms coincide. Second, the actuator forces of the reference model enter the chain dynamics as *external* wrenches: the machine frame does not feel the stage-acceleration reaction force, which in the physical machine is absorbed by the balance mass of each chain. Since MuJoCo joint actuators react on the parent body, the simulation emulates the balance masses by cancelling the long-stroke actuator reaction on the base frame at every integration step. Without this cancellation the 1400 kg base frame recoils by close to a millimetre during the step-and-scan cycle and its ringing, fed back into both chains, dominates the exposure error.

The simulation incorporates a *StepperController* that generates the references for the scanning and stepping phases. During the scanning phase, the wafer stage moves at a constant velocity $v_{w,scan} = 0.8$ m/s while the reticle stage moves in the opposite direction at $v_{r,scan} = 3.2$ m/s, maintaining the demagnification ratio $\alpha = 0.25$. This testing framework allows us to evaluate the impact of the centripetal and Coriolis terms from Eq. (2.2.11) and the effectiveness of the inertia aggregation described in Algorithm 1 when subjected to high-acceleration ($a_{w,max} = 20$ m/s^2) transients.

In addition to the full multi-body simulation, a reduced-order representation of the machine is used when long batches of runs are required. Following the Case 1 configuration of the characterization study[2], each chain collapses to two bodies: an actuated *coarse stage* carrying the position encoder, and a passive *end-effector* (the wafer or mask holder) coupled to it through the corresponding Kelvin–Voigt element. The resulting system of ordinary differential equations is integrated with an adaptive Runge–Kutta scheme at a maximum step of 10^{-4} s. This reduced model preserves the dominant two-mass resonance of each chain, which governs the synchronization error during exposure, at a small fraction of the computational cost of the full model.

4.3 Reference Trajectory Generation

The trajectories commanded to the stages are **fourth-order profiles**: piecewise-polynomial motions in which position, velocity, acceleration, and jerk are all continuous, and the fourth derivative — the *snap* — switches between $\pm s_{\max}$ and zero[11]. Within each segment the position evolves as

$$p(t) = p_0 + v_0 t + \frac{a_0}{2} t^2 + \frac{j_0}{6} t^3 + \frac{s}{24} t^4, \qquad (4.3.1)$$

where (p_0, v_0, a_0, j_0) are the boundary conditions inherited from the previous segment and $s \in \{-s_{\max}, 0, +s_{\max}\}$. A symmetric point-to-point motion consists of fifteen such segments, arranged so that none of the kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ is exceeded.

Two generators are implemented in this work, and the distinction matters for interpreting the experiments. The *first* is a polynomial approximation: the acceleration ramp is a sixth-order smoothstep whose duration is computed from $(v_{\max}, a_{\max}, j_{\max})$ and the travel distance. Its motion is continuous through jerk with bounded snap, but the snap bound is *implied* by the ramp duration ($s_{peak} = 60\, v_{\max}/t_{accel}^3$) rather than imposed by $s_{\max}$; the pipeline experiments of Chapter 5 that predate this observation (Experiments 3 and 5) use this generator, whose effective parameterization is $(v_{\max}, a_{\max}, j_{\max})$.

The *second* is the exact generator: a segment-planning implementation that constructs the profile from piecewise-constant snap — seven segments per acceleration ramp, fifteen in total — with the feasibility cascade of[11]: if the jerk plateau is unreachable the effective jerk reduces to $\sqrt{a_{\max} s_{\max}}$, if the acceleration plateau is unreachable the effective acceleration follows from the corresponding quadratic, and if the travel distance cannot accommodate the ramps the cruise velocity is reduced by bisection. The same machinery instantiates the lower-order profiles used for comparison: a trapezoidal acceleration-bounded profile (second order) and a jerk-bounded S-curve (third order, seven segments). All generators were verified over a grid of parameter combinations, including degenerate short-move and tight-snap cases, for exact travel distance, terminal rest, and respect of every kinematic bound. The trajectory-order study and the optimization scenarios (Experiments 6 and 7) use the exact generators, restoring $s_{\max}$ as a live parameter of the search space. All profiles expose the full kinematic state (p, v, a, j, s) at any time instant.

The order of the profile determines the high-frequency content of the force commanded to the stage. A second-order (trapezoidal-velocity) profile demands instantaneous jumps in acceleration, injecting energy across a broad frequency band; a third-order profile bounds the jerk but still steps it discontinuously[4]. Bounding the snap rolls off the excitation spectrum at $-80\,\mathrm{dB}$ per decade, which suppresses the response of poorly damped structural modes — such as the reticle end-effector resonance identified in Chapter 2 — and shortens the residual settling tail after each step[11;1]. The scanning phase itself takes place inside the constant-velocity segment of the profile, where ideally no force beyond the viscous terms is required.

4.4 Stage Feedback Control

The control architecture is organized as a ladder of configurations of increasing capability, mirroring the case structure of the characterization study[2]. Each configuration adds one mechanism on top of the previous, so that the contribution of every element to the exposure indices can be isolated experimentally in Chapter 5.

Case 1 — collocated long-stroke control. Each coarse stage is closed with a collocated PID controller with inertia feedforward,

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d\, \dot{e}(t) + m\, \ddot{r}(t), \tag{4.4.1}$$

where $e(t) = r(t) - y_c(t)$ is the error measured at the coarse-stage encoder, $\ddot{r}(t)$ is the reference acceleration supplied by the trajectory generator, and m is the total moving mass of the chain ($\approx 87.7\,\mathrm{kg}$ for the wafer chain). The gains are designed for a bandwidth of $f_{bw} = 50\,\mathrm{Hz}$ following the loop-shaping rules of[3], giving $K_p \approx 8.66 \times 10^6\,\mathrm{N/m}$, $K_i \approx 2.72 \times 10^8\,\mathrm{N/(m{\cdot}s)}$, and $K_d \approx 2.75 \times 10^4\,\mathrm{N{\cdot}s/m}$, with the actuator force saturated at $5 \times 10^4\,\mathrm{N}$. Feedback is taken from the coarse stage rather than the end-effector because non-collocated control through the lightly damped reticle coupling ($\zeta \approx 0.04$) destabilizes the loop; the price of collocation is that the lag between the coarse stage and the end-effector remains uncompensated and appears directly in the synchronization error. In the full simulation model this lag is dominated by its acceleration-proportional component, with two contributions: the servo deflection $m\,a/K_P$ of the position loop and the quasi-static stretch $m_d\, a/K$ of each passive coupling carrying the downstream mass m_d; on the frictionless plant the velocity-proportional component is negligible.

Case 2 — lag-compensation feedforward. Since both lag components are functions of the known reference kinematics, they can be cancelled by commanding the long stroke *ahead* of the reference:

$$r_{LS}(t) = r(t) - \big(\ell_v\, \dot{r}(t) + \ell_a\, \ddot{r}(t)\big), \tag{4.4.2}$$

so that the end-effector, not the coarse carrier, lands on the commanded position. The coefficients (ℓ_v, ℓ_a) of each chain are identified from the simulation itself by iterative least squares: the sequence is run, the measured end-effector lag is regressed on the reference velocity and acceleration, the fit is added to the feedforward, and the procedure repeats on the residual. Iteration is required because the lag model is quasi-static: the decaying ringing of the flexible couplings after each acceleration ramp biases a single pass, and three passes reduce the residual to its converged value. What the fit cannot represent — and what remains after Case 2 — is precisely that dynamic transient, which motivates the fine-stage feedback below. This is the model-free counterpart of the spring-compensation configuration of the reference study, and requires no additional hardware or sensing.

Cases 3a/3b — fine-stage feedback. The remaining error after feedforward is non-repeatable transient response, which only feedback can remove. The fine stage of each chain closes a position loop on the measured end-effector error $e_f(t)$ (in production machines this measurement is provided by the interferometry that also serves the exposure metrology). Case 3a uses a linear PI law with anti-windup. Case 3b replaces the linear integrator with a **hybrid integrator–gain system** (HIGS)[6]: an integrator constrained to operate in the sector where its state x_1 and its input agree in sign,

$$\dot{x}_1 = \omega_h\, e_f \quad \text{while } e_f\, x_1 \geq x_1^2/k_h, \qquad x_1 = k_h\, e_f \quad \text{otherwise}, \tag{4.4.3}$$

with output $u_f = k_p e_f + \omega_i \int x_1\, dt$. By resetting its state to the gain line $k_h e_f$ whenever the sector condition is violated, the HIGS element provides integral action with roughly 38° less phase lag than a linear integrator, relaxing the overshoot-versus-stiffness limitation of linear designs — the property that motivates nonlinear integrators for wafer stages. Both fine loops saturate at the physical travel of the short stroke, with conditional integration preventing windup during large transients. The loop gains are tuned empirically over a grid up to the stability boundary of the simulated plant: the PI loop uses $k_p = 40$, $k_i = 15{,}000\,\mathrm{s}^{-1}$, and the HIGS loop $k_p = 40$, $\omega_i = 2\pi \cdot 2500\,\mathrm{rad/s}$, $\omega_h = 2\pi \cdot 1000\,\mathrm{rad/s}$, $k_h = 1$; one step further in gain destabilizes both loops against the flexible-chain modes. The HIGS element sustains an integrator frequency two orders of magnitude above the PI's equivalent k_i/k_p before reaching that boundary — the phase advantage of the nonlinear integrator expressed as usable gain. The fine-stage authority is ultimately bounded by its own actuator velocity loop (pole at $K_P/K_V = 100\,\mathrm{rad/s}$), a structural property of this generic replication that Chapter 5 quantifies.

4.5 Population-Based Trajectory Search

The optimization problem (3.4.2) is searched with particle swarm optimization (PSO) over the four kinematic bounds. Each particle holds a position $\mathbf{x} = (v_{\max}, a_{\max}, j_{\max}, s_{\max})$ and is updated by the standard rule

$$\mathbf{u}_{k+1} = \omega\,\mathbf{u}_k + c_1 r_1(\mathbf{p}_k - \mathbf{x}_k) + c_2 r_2(\mathbf{g}_k - \mathbf{x}_k), \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}_{k+1}, \tag{4.5.1}$$

with inertia $\omega = 0.6$, cognitive and social weights $c_1 = c_2 = 1.4$, and positions clipped to the search box. Evaluating one particle runs the full step-and-scan sequence over the wafer under the candidate trajectory, converts the per-die exposure indices into a wafer map, and prices the map. Two cost definitions are compared as the scenarios E3 and E4 of the evaluation plan: a conventional servo cost (the wafer-average exposure MSD, normalized by the failure threshold) and the yield-aware cost built on the CNN. For the latter, the weighted form (3.4.1) is instantiated as

$$\mathcal{J}_{map} = \bigl(1 - P_{none}\bigr) + \sum_{c \in \mathcal{C}_{ood}} P_c, \tag{4.5.2}$$

where $\mathcal{C}_{ood}$ is the set of classes that scanner motion cannot plausibly produce. The first term prices every predicted defect; the second penalizes maps whose morphology the classifier attributes to non-scanner mechanisms, closing the blind spot identified in Chapter 5 where saturated maps concentrate probability outside the plausible subset and a plausible-classes-only cost would vanish. Both scenarios share a soft throughput constraint, $\mathcal{J} = \mathcal{J}_{map} + \beta \max(0, T_{wafer} - T_{budget})/T_{budget}$ with $\beta = 4$, where T_{wafer} follows from the scan-profile duration, the settling budget, and the die count.

4.6 From Synchronization Error to Wafer Maps

The link between trajectory parameters and yield is established in three steps: the simulated synchronization error is aggregated per die, the resulting pass/fail pattern forms a simulated wafer map, and a convolutional neural network classifies that map into a defect-morphology class.

Dataset. The CNN is trained on WM-811K[16], a public dataset of 811,457 wafer maps collected from real fabrication lines, of which 172,950 carry a manually assigned failure label. Each map is a grid of dies marked as pass or fail, annotated with one of nine morphology classes: *none*, *Center*, *Donut*, *Edge-Loc*, *Edge-Ring*, *Loc*, *Near-full*, *Random*, and *Scratch*. The labelled maps are split 80/20 with class stratification, and every map is resized to 64×64 cells to give the network a uniform input.

Architecture and training. The network comprises three convolutional blocks (16, 32, and 64 filters of size 3×3, each followed by ReLU activation and 2×2 max-pooling) and two fully connected layers (128 hidden units with dropout of 0.5, and a 9-way softmax output). It is trained for 5 epochs with the Adam optimizer (learning rate 10^{-3}, batch size 32) under a cross-entropy loss, retaining the weights with the highest hold-out accuracy.

Simulated maps. For each die scanned in simulation, the synchronization error recorded during its exposure is reduced to a scalar criterion — either the maximum tracking error or the MA/MSD pair of Chapter 3 — and compared against a threshold to declare the die good or failing. The resulting grid, arranged in the same encoding as WM-811K (background, pass, fail), is resized to 64×64 and presented to the CNN, which returns the probability distribution P_c used by the cost function (3.4.1).

Validation criterion. A physically meaningful trajectory-to-defect mapping must activate only the classes plausibly caused by scanner motion: *none*, *Scratch*, *Edge-Loc*, and *Random*. The remaining classes correspond to etching, cleaning, and deposition signatures; if simulated maps systematically activated them, the mapping — not the machine — would be at fault. This criterion, together with the requirement that the cost shift reproducibly under sweeps of $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$, defines what constitutes a successful validation in Chapter 5.

4.7 Recurrent Surrogate Model

Evaluating the cost of a single candidate trajectory requires a full multi-body simulation, which makes population-based optimization over $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ expensive. To accelerate the search, a recurrent surrogate model is trained to predict the synchronization error signal directly from the kinematic profile. The surrogate is a two-layer gated recurrent unit (GRU) network with 64 hidden units and dropout of 0.2, mapping the instantaneous kinematic state (v, a, j, s) of the reference to the synchronization error e_{syn} sample by sample. Its training set is generated by running the simulation over randomized trajectory parameters — $v \in [0.01, 1.0]\,\mathrm{m/s}$, $a \in [0.1, 40]\,\mathrm{m/s^2}$, $j \in [10, 5000]\,\mathrm{m/s^3}$, $s \in [10^2, 5 \times 10^5]\,\mathrm{m/s^4}$ — and recording the error during the scanning phases. One hundred randomized runs yield approximately 1.6 million training samples, split 80/20 by simulation so that no trajectory contributes to both sets. Features and targets are standardized, and the network is trained for 10 epochs with Adam (learning rate 10^{-3}, batch size 32) under a mean-squared-error loss. Once trained, the surrogate predicts the MA/MSD of a candidate trajectory in milliseconds, allowing exhaustive or population-based searches over the parameter space before any promising candidate is verified in the full simulation.

A scope remark is in order. For a purely linear plant under linear control, this map could be computed exactly by classical means, and a learned model would be redundant; the surrogate is introduced here because the target configurations of the framework are not in that regime — actuator saturation, the nonlinear feedback structures planned for the fine-stage machine, and full-wafer simulations whose per-candidate cost dominates the optimization loop. The conditions under which the recurrent surrogate is and is not justified are examined critically, against the experimental evidence, in Section 5.9.1.

Chapter 5

Experiments and Results

5.1 Introduction

This chapter reports the experimental campaign carried out over the simulation framework of Chapter 4. Experiment 1 validates the simulated machine against the published Case 1 study it derives from, establishing the dynamic behavior of the plant without fine stages. Experiment 2 trains and evaluates the convolutional network on the WM-811K dataset, establishing the classifier half of the pipeline. Experiment 3 connects the two: it verifies that disturbances injected into the simulation produce wafer-map signatures the CNN can react to, and that sweeping the kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ shifts the exposure error — and therefore the defect map — reproducibly. Experiment 4 climbs the control ladder of Section 4.4: lag-compensation feedforward, fine-stage PI, and the nonlinear HIGS loop, quantifying the contribution of each. Experiment 5 is the central demonstration of the thesis premise: as the settling budget shrinks, trajectory-induced failures emerge as a spatially structured, scanner-plausible pattern, and the yield-aware cost traces the throughput–yield transition. Experiment 6 runs the optimization scenarios E3 and E4, comparing the conventional servo cost against the CNN cost inside the same particle-swarm search over the exact snap-bounded profile family. Experiment 7 crosses profile order with controller quality, quantifying whether trajectory smoothness matters when the controller is poor and how much it is worth when the controller is good. A closing secondary study evaluates the recurrent surrogate model. All results reported below are outputs of the actual scripts and trained models of this project; where a result falls short of the criterion set in Chapter 3, this is stated and analyzed rather than omitted.

5.2 Experiment 1: Reproduction of the Case 1 Study

The first experiment reproduces Case 1 of the characterization study by Al-Rawashdeh et al.[2]: planar step-and-scan motion of the wafer and reticle chains *without* fine stages, under the collocated PID-plus-feedforward controller of Section 4.4. The purpose is twofold: to verify that the simulation reproduces the qualitative conclusions of the reference study, and to quantify how far a coarse-only machine falls from the exposure specifications, which motivates everything that follows.

The reduced two-body-per-chain model is integrated over a four-die exposure sequence (dies of $32 \times 32\,\mathrm{mm}^2$ in one row, snake pattern) with the baseline trajectory parameters $v_{w,scan} = 0.8\,\mathrm{m/s}$, $a_{w,\max} = 20\,\mathrm{m/s}^2$, $j_{\max} = 1600\,\mathrm{m/s}^3$: the scan profile lasts 145 ms with a constant-velocity cruise between 52.5 and 92.5 ms, each step takes 82.2 ms, and the whole sequence spans 1.727 s. The reticle moves at 3.2 m/s with $a_{r,\max} = 80\,\mathrm{m/s}^2$ in the opposite direction. Figure 5.1 shows the commanded references, and Figure 5.2 the resulting tracking and synchronization errors with the exposure windows highlighted.

The dynamic structure of the simulated machine matches the reference study. The wafer chain exhibits its two-mass resonance at 190.3 Hz with a well-damped coupling ($\zeta = 0.70$), while the reticle chain resonates at 131.4 Hz with $\zeta = 0.044$: the reticle end-effector is lightly damped, and non-collocated feedback through this coupling is unstable, which forces the collocated architecture and leaves the coupling lag uncompensated. During the acceleration ramps this lag reaches its quasi-static value $\Delta = m_e a/K$: 15.9 µm for the wafer end-effector (10.7 kg at

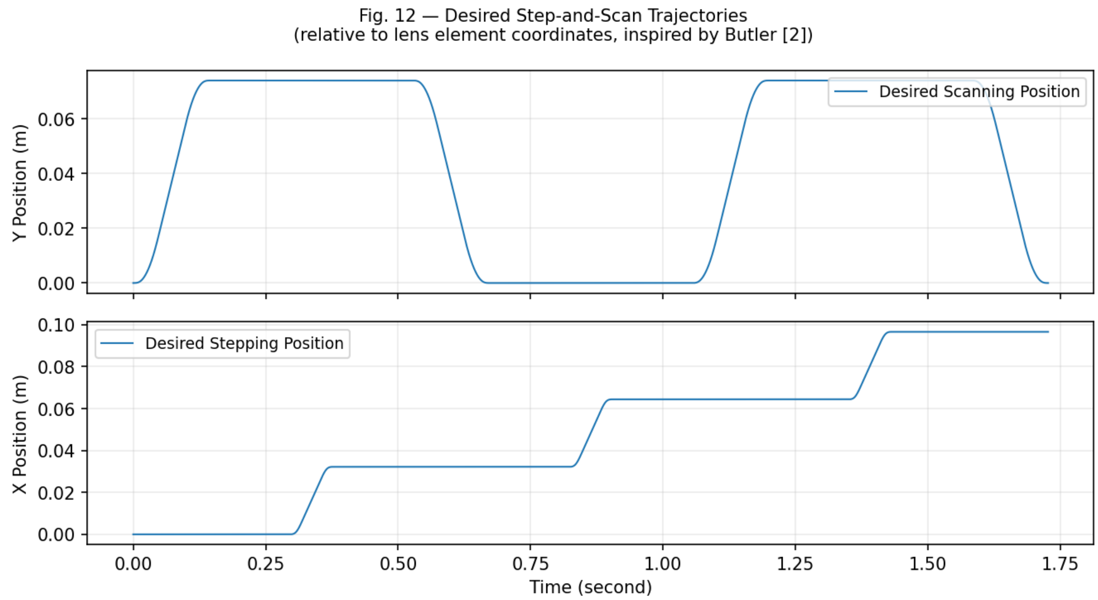


Figure 5.1: Commanded step-and-scan references for the wafer stage (x: stepping, y: scanning) over the four-die sequence.

$20\,\mathrm{m/s^2}$) and 133.5 µm for the reticle end-effector (10.54 kg at $80\,\mathrm{m/s^2}$), leaving a net quasi-static synchronization error of about $-17.4\,\mu\mathrm{m}$ during the ramps (Figure 5.3).

Inside the exposure windows — where only the residual oscillation of the lightly damped reticle mode remains — the performance indices evaluate to

$$\mathrm{MA}_{syn} \in [-985.3,\ +985.2]\ \mathrm{nm}, \qquad \max \mathrm{MSD}_{syn} = 427.8\ \mathrm{nm}, \tag{5.2.1}$$

against the specifications $\mathrm{MA} \in [-1.25, +2]\,\mathrm{nm}$ and $\mathrm{MSD} \leq 7\,\mathrm{nm}$ of Chapter 3. Figure 5.4 shows both indices over the sequence and Figure 5.5 the detail of the first exposure window.

The conclusion of the experiment is the same as that of the reference study: a machine without fine stages misses the exposure specification by nearly three orders of magnitude, with the error dominated by the residual 131 Hz oscillation of the reticle end-effector excited during the acceleration ramps. This is the error mechanism that trajectory shaping acts upon — gentler kinematic bounds inject less energy at the resonance — and the mechanism that the remaining experiments quantify at the wafer-map level.

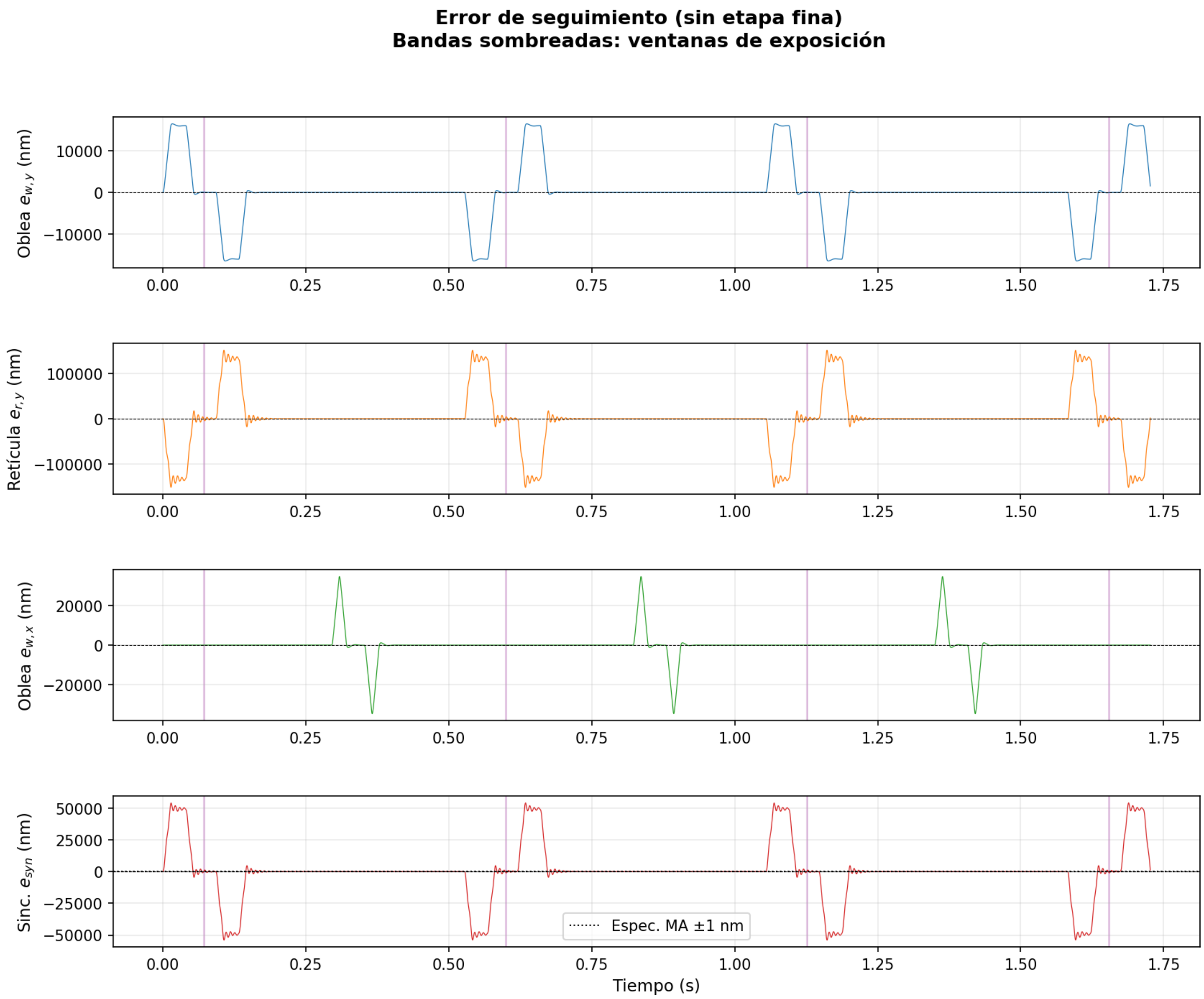


Figure 5.2: Wafer, reticle, and synchronization errors during the four-die sequence. Shaded bands mark the exposure windows.

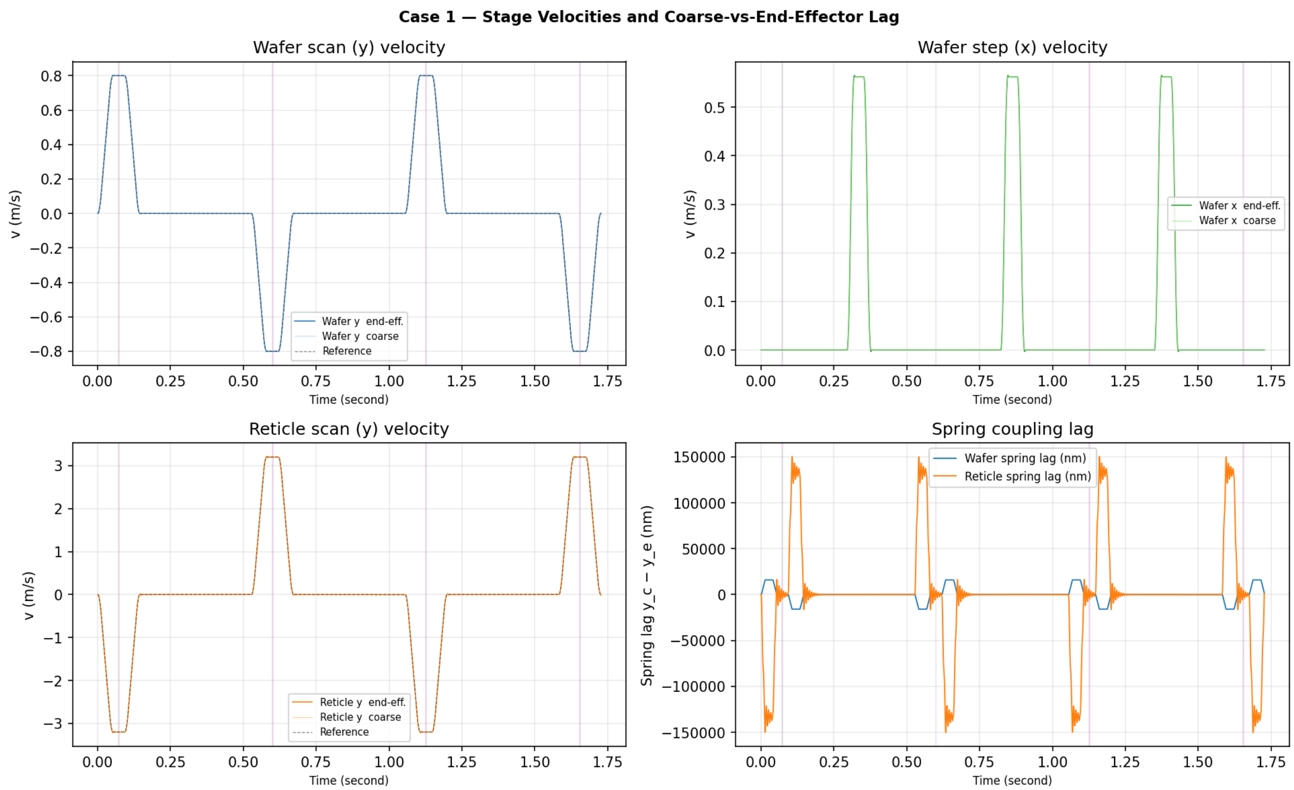


Figure 5.3: Stage velocities and spring-coupling lag during the scan. The reticle end-effector, four times faster and an order of magnitude more softly coupled, dominates the synchronization error.

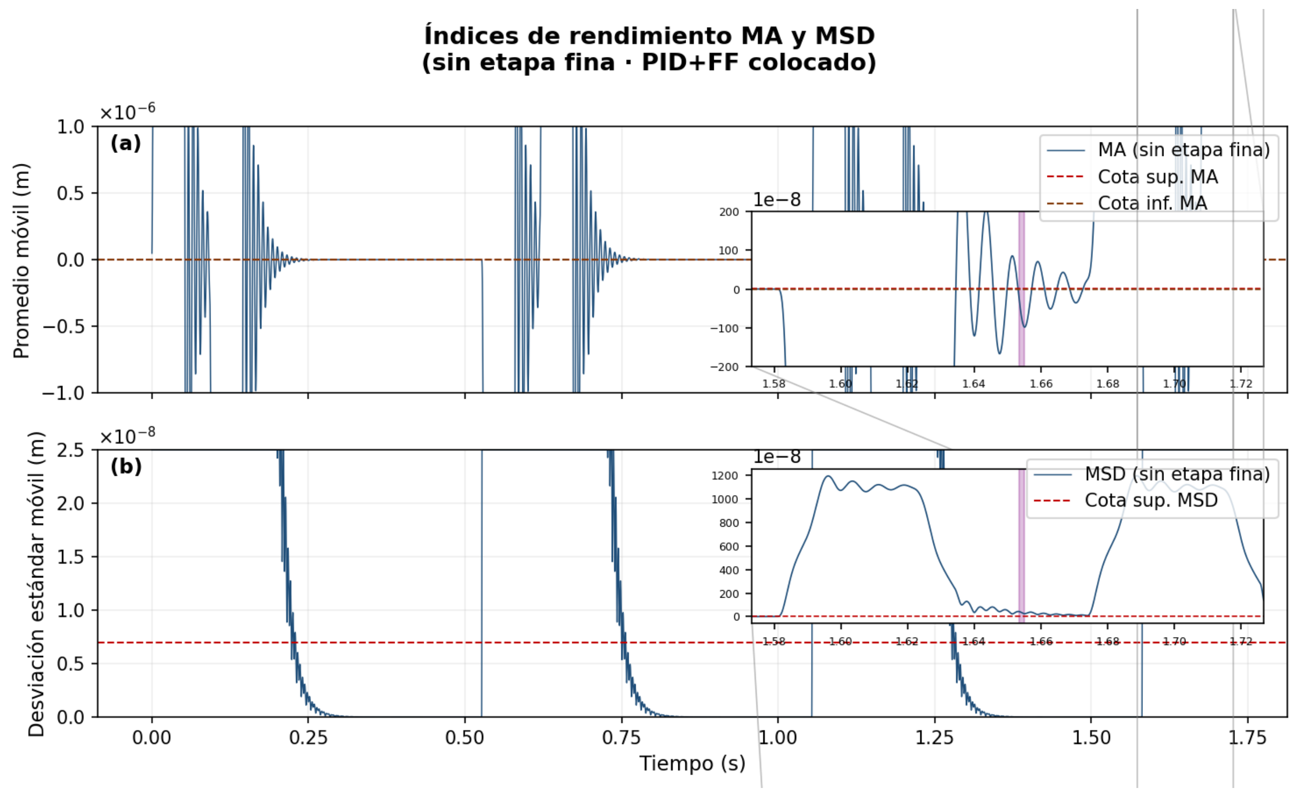


Figure 5.4: MA and MSD performance indices of the synchronization error over the four-die sequence.

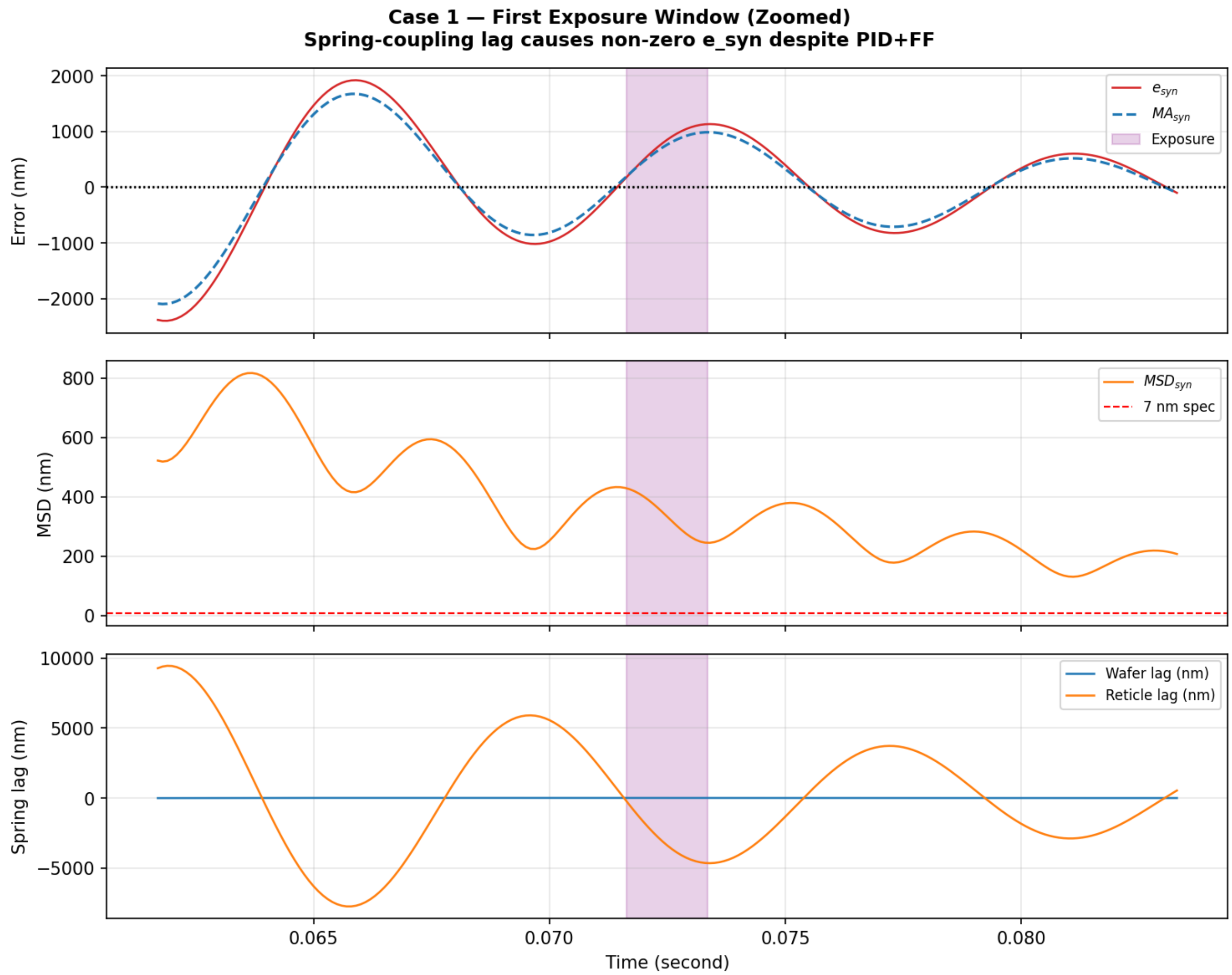


Figure 5.5: Detail of the first exposure window: synchronization error, MA, MSD, and coupling lag.

5.3 Experiment 2: CNN Training and Evaluation on WM-811K

The second experiment trains the classifier of Section 4.6 and evaluates it on held-out data. Of the 811,457 maps in WM-811K, the 172,950 maps carrying a failure label are used; the stratified 80/20 split leaves 34,590 maps for evaluation, never seen during training.

The network reaches an overall hold-out accuracy of **96.88 %** with a macro-averaged F1 score of 0.77. Table 5.3.1 reports the per-class precision, recall, and F1, and Figure 5.6 the row-normalized confusion matrix.

Table 5.3.1: Per-class evaluation of the CNN on the WM-811K hold-out set (34,590 maps).

Class	Precision	Recall	F1	Support
none	0.98	1.00	0.99	29,486
Center	0.95	0.90	0.92	859
Donut	0.85	0.84	0.85	111
Edge-Loc	0.87	0.75	0.80	1,038
Edge-Ring	0.94	0.99	0.97	1,936
Loc	0.76	0.63	0.69	718
Near-full	0.96	0.80	0.87	30
Random	0.87	0.76	0.81	173
Scratch	0.00	0.00	0.00	239
accuracy		0.9688		
macro avg	0.80	0.74	0.77	34,590

The matrix is strongly diagonal for eight of the nine classes. The exception is *Scratch*, whose recall is zero: with 239 test samples against 29,486 *none* samples, the cross-entropy objective is

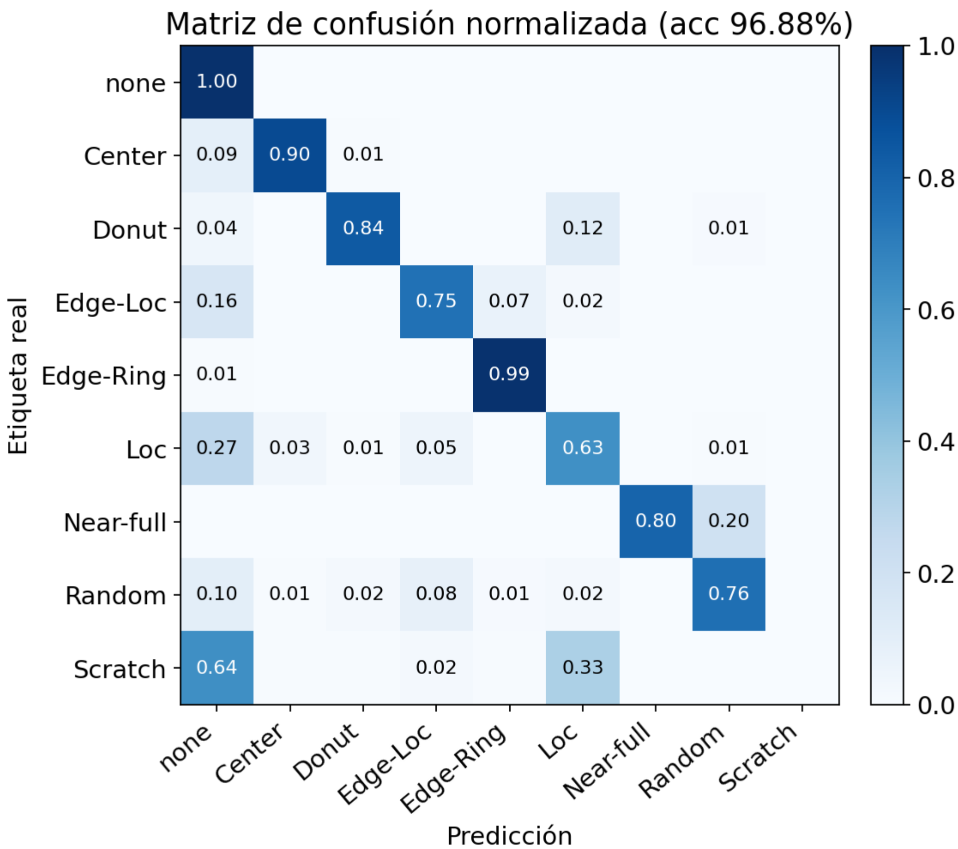


Figure 5.6: Row-normalized confusion matrix of the CNN on the WM-811K hold-out set.

dominated by the majority classes and the network learns to absorb the thin-line morphology into neighboring classes. This is a known consequence of the severe class imbalance of WM-811K and matters here because *Scratch* is one of the four scanner-plausible classes. Mitigation — class-weighted loss, oversampling, or focal loss — is deliberately deferred: it modifies only the training recipe of Experiment 2 and leaves the rest of the pipeline unchanged.

5.4 Experiment 3: Trajectory-to-Defect Pipeline

The third experiment exercises the full bridge of Section 4.6 over the complete 300 mm wafer (49 dies of $32 \times 32\,\mathrm{mm}^2$) in the full MuJoCo model. Two campaigns are run: disturbance injection at fixed trajectory parameters, and a sweep of the kinematic bounds at fixed (zero) disturbance. Both campaigns operate under Case 1 control (long-stroke tracking only); Experiment 4 then upgrades the controller, and Experiments 5 and 6 run on top of the upgraded configuration.

A practical observation conditions both campaigns. In the Case 1 machine, the exposure-window MSD of the synchronization error sits between 45 and 390 µm depending on the trajectory — four orders of magnitude above the 7 nm specification, consistently with Experiment 4's ladder baseline. An absolute threshold at the specification therefore marks every die of every wafer as failing and produces saturated, uninformative maps. Both campaigns consequently use criteria referenced to the unperturbed baseline of the same machine: what is measured is the *relative* effect of the disturbance or of the trajectory parameters, not compliance with a specification that this machine configuration cannot reach.

5.4.1 Disturbance signatures

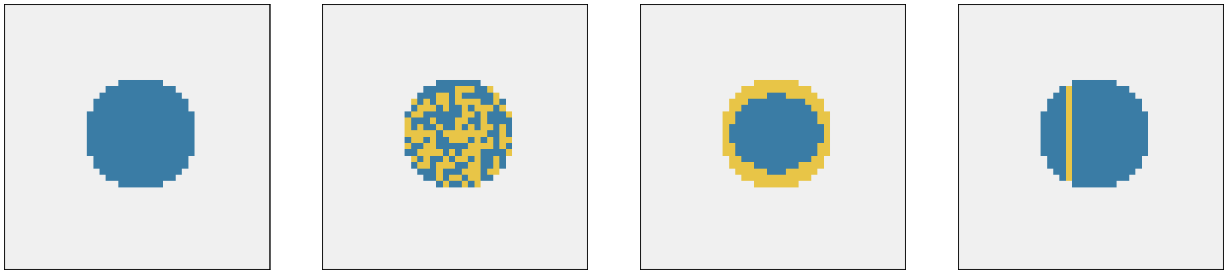
Three disturbance types are injected as forces on the wafer end-effector while the baseline trajectory runs: broadband force noise on both axes (floor *vibration*), a sustained bias force active while scanning dies near the wafer edge (*drift*), and a sustained bias force active while scanning a single die column (*scratch*). For every die, the synchronization error during the central third of the scan — the constant-velocity cruise that contains the exposure — is reduced to its mean and its maximum moving standard deviation. A die is marked as failing when its mean shifts by more than 100 nm with respect to the same die in the unperturbed run, or when its MSD exceeds twice the baseline median; both criteria are immune to the constant geometric offset of the absolute coordinates. The resulting maps are classified by the CNN of Experiment 2.

This campaign uses a finer die grid than the sweep — $16 \times 16\,\mathrm{mm}^2$ dies, 241 per wafer — so that line- and point-like patterns retain their morphology after resizing to the CNN input. Table 5.4.2 and Figure 5.7 present the results.

Table 5.4.2: Disturbance-injection campaign over 241 dies. Expected class refers to the scanner-plausible morphology the disturbance emulates.

Disturbance	Failing dies	Spatial pattern	Expected class	CNN prediction
none	0/241	clean wafer	none	none (97.2 %)
vibration	114/241	scattered	Random	Center (88.6 %)
drift	100/241	outer ring	Edge-Ring / Edge-Loc	Donut (99.9 %)
scratch	15/241	one die column	Scratch	Loc (63.1 %)

The spatial half of the pipeline behaves as designed: the unperturbed wafer is clean and each disturbance produces a failing-die population of the expected shape and location — scattered dies under force noise, an outer ring under the edge bias, and a single 15-die column under the column bias. The classification half is only partially aligned with the scanner-plausible target classes. The clean wafer is correctly assigned to *none*, satisfying the first requirement of the validation criterion. The remaining assignments land on morphological neighbors rather than on the intended classes: the ring is read as *Donut* — a defensible reading, since the drift band covers an annulus wider than the one-die rim that characterizes *Edge-Ring* in WM-811K —



(a) none (b) vibration (c) drift (d) scratch

Figure 5.7: Simulated wafer maps under the four disturbance conditions (yellow: failing dies, blue: passing dies). The spatial signatures — clean, scattered, ring, line — correspond to the injected disturbances.

the column is read as *Loc*, the closest available class given that the classifier's *Scratch* recall is zero (Experiment 2), and the scattered pattern, at a density of 47 % of the wafer, is absorbed into *Center* rather than *Random*. These deviations trace back to the classifier's minority-class weakness and to the density of the injected failures, not to the simulation half of the pipeline; they are taken up in the discussion of Section 5.10.

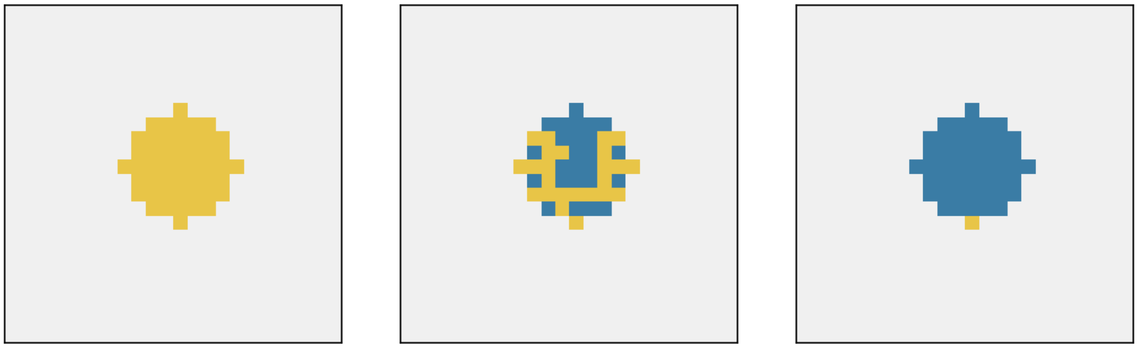
5.4.2 Kinematic-bound sweep

The second campaign fixes the disturbances at zero and sweeps the kinematic bounds around the baseline (0.8 m/s, 20 m/s^2, 1600 m/s^3, 10^5 m/s^4): one aggressive configuration raises all four bounds, and four configurations relax one bound at a time; a conservative configuration relaxes all four. For every die the maximum moving-window MSD of the synchronization error during the scan is computed, a die fails when it exceeds the scaled threshold (the baseline median, 152 µm), and the map is classified by the CNN; the cost is the total probability assigned to the scanner-plausible defect classes.

Table 5.4.3 summarizes the seven configurations. The median exposure MSD responds reproducibly to every kinematic bound, and monotonically to the force-shaping ones: raising all bounds multiplies it by 2.5 (152 to 388 µm), while halving the acceleration or quartering the jerk reduces it to 87 and 52 µm respectively, with the conservative configuration reaching 45 µm. The velocity bound behaves differently on the frictionless plant: halving v leaves the median slightly *higher* (160 µm) and fails 44 of 49 dies, because the exposure window of a point on the wafer scales as $1/v$ — a slower scan integrates more of the decaying post-ramp transient into every die's MSD. Aggressiveness in the force-shaping bounds (a, j) injects more transient; slowness in v exposes each point to it for longer; the optimizer of Experiment 6 must balance the two, and indeed settles above the baseline velocity. Tightening the snap bound by a factor of ten changes nothing — profile duration and every error statistic are bit-for-bit identical to the baseline. This experiment is in fact what exposed the implementation property documented in Section 4.3: the polynomial generator's snap is implied by the ramp duration, so the $s_{\max}$ parameter is inert and the effective search space of all optimization experiments is $(v_{\max}, a_{\max}, j_{\max})$. Since the runs are deterministic, repeated execution reproduces these values exactly, satisfying the reproducibility half of the validation criterion.

Table 5.4.3: Kinematic-bound sweep over the full 49-die wafer. T_{scan} is the duration of one scan profile; the failure threshold is the baseline median MSD (152 µm); cost is the total CNN probability of the scanner-plausible defect classes.

Config	v [m/s]	a [m/s^2]	j [m/s^3]	s [m/s^4]	T_{scan} [ms]	med. MSD [µm]	Failing dies	CNN top class (prob.)
Aggressive	1.2	40	5000	5×10^5	102.7	388	49/49	Center (1.000)
Baseline	0.8	20	1600	10^5	145.0	152	24/49	Center (0.973)
Tighter snap	0.8	20	1600	10^4	145.0	152	24/49	Center (0.973)
Half velocity	0.4	20	1600	10^5	145.0	160	44/49	Center (1.000)
Half accel.	0.8	10	1600	10^5	212.5	87	1/49	none (0.991)
Quarter jerk	0.8	20	400	10^5	504.0	52	1/49	none (0.991)
Conservative	0.2	5	200	5×10^3	290.0	45	1/49	none (0.991)



(a) aggressive (b) baseline (c) conservative

Figure 5.8: Simulated wafer maps under the scaled MSD criterion for three points of the sweep (yellow: failing dies, blue: passing dies). The failing-die population collapses from full-wafer to a single die as the kinematic bounds are relaxed.

The wafer maps (Figure 5.8) and their classification expose a property of the cost function that matters for the optimization stage. Configurations that pass have $P_{none} = 0.991$ and a residual defect-class cost of 0.0056; but the saturated maps of the aggressive and baseline configurations concentrate essentially all probability in *Center* — a class outside the scanner-plausible subset — so the cost restricted to plausible classes evaluates to zero exactly where the wafer is worst. A dense, wafer-wide failure pattern has no thin-line or scattered morphology left for the CNN to recognize. For the optimizer of Eq. (3.4.2) this dictates either complementing the cost with the $1 - P_{none}$ term or penalizing any activation of the non-plausible classes as an out-of-domain signal; both preserve the CNN as the cost backbone while removing the blind spot.

5.5 Experiment 4: Climbing the Control Ladder

The pipeline experiments above ran under Case 1 control, where the exposure error is millimetre-scale and the failure thresholds had to be scaled accordingly. This experiment adds, one at a time, the mechanisms of Section 4.4 that make fourth-order trajectories trackable, and measures the contribution of each on the baseline trajectory over a four-die row in the full simulation model. Table 5.5.4 reports the cruise-window indices of the offset-corrected synchronization error; Figure 5.9 shows the error traces.

Table 5.5.4: Controller ladder on the baseline trajectory (cruise-window indices of e_{syn}, worst die).

Configuration	max |MA|	max MSD
Case 1: long-stroke PID only	737.9 µm	33.9 µm
Case 2: + lag-compensation FF	180.9 µm	11.0 µm
Case 3a: + fine-stage PI	2.50 µm	0.22 µm
Case 3b: + fine-stage HIGS	2.14 µm	0.27 µm

Three observations follow. First, the identified lag feedforward reduces the exposure-window MA by a factor of four: on the frictionless plant the dominant Case 1 error is the acceleration-proportional lag of the chain — servo deflection plus coupling stretch, largest on the reticle chain at $80\,\mathrm{m/s^2}$ — and its quasi-static part is cancelled open-loop by commanding the long stroke ahead of the reference (Eq. (4.4.2)); what survives the fit is the *dynamic* part, the decaying ringing of the flexible couplings after each ramp, which no quasi-static feedforward can represent. Second, the fine-stage loops attack exactly that transient and reduce the exposure MSD by a further factor of 40–50 (11.0 µm to 0.22–0.27 µm), a combined improvement of 345× on MA and 157× on MSD over Case 1; the residual MA of 2.1–2.5 µm is the tail of the post-ramp transient still decaying at cruise entry, and it is bounded by the fine loop's authority — the fine-stage actuator's velocity loop places its command-following pole at $K_P/K_V = 100\,\mathrm{rad/s}$, and gain beyond $k_p \approx 60$–80 destabilizes the loop against the flexible-chain modes. Third, at their respective stability-boundary tunings the linear PI and the nonlinear HIGS loop perform within 25 % of each other on this metric (HIGS is 14 % better on MA, PI 25 % better on MSD), although the HIGS element sustains an integrator frequency two orders of magnitude above the PI's equivalent before destabilizing — its documented phase advantage expressed as usable gain; the

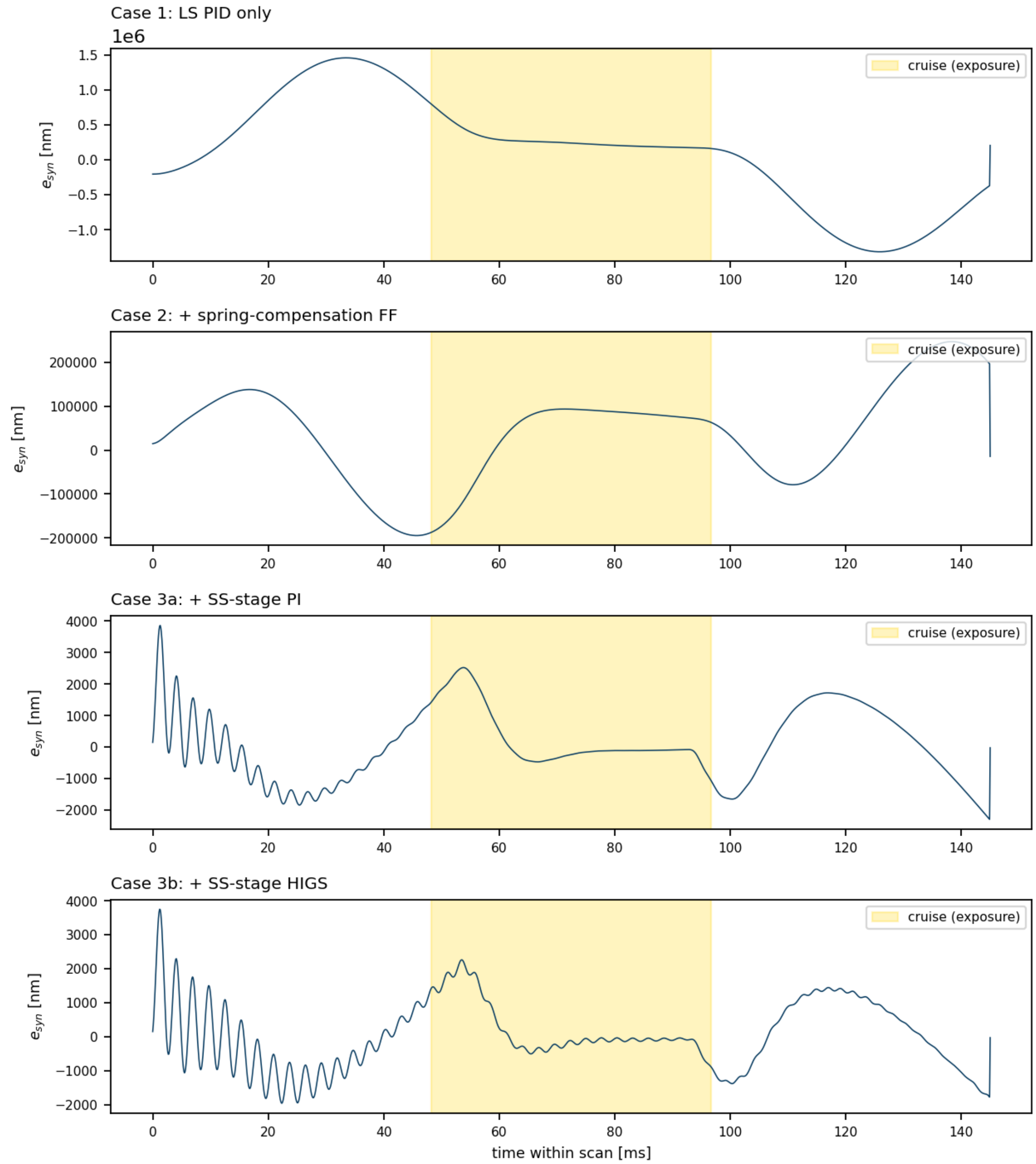


Figure 5.9: Synchronization error during one scan under the four control configurations. Shaded band: cruise (exposure) segment.

regime where that advantage becomes decisive, overshoot-constrained settling, is not isolated by the present cruise-tracking metric. The nanometre specifications themselves remain out of reach by three further orders of magnitude, and the remaining gap is now attributable to the fine stage rather than to the plant model: the reference machine reaches its nanometre synchronization band with a purpose-designed fine positioning stage whose resonances sit at 2 kHz and whose actuator mounts are essentially rigid (10^{12} N/m in the published parameter set[2]), five orders stiffer than the 10^{7} N/m servo of this generic replication. All subsequent experiments therefore run under Case 3b and reference their thresholds to its baseline, studying the *relative* effect of trajectory parameters at the error scale this plant provides — now sub-micrometre rather than the millimetre scale of the uncorrected plant.

5.6 Experiment 5: The Throughput–Yield Transition

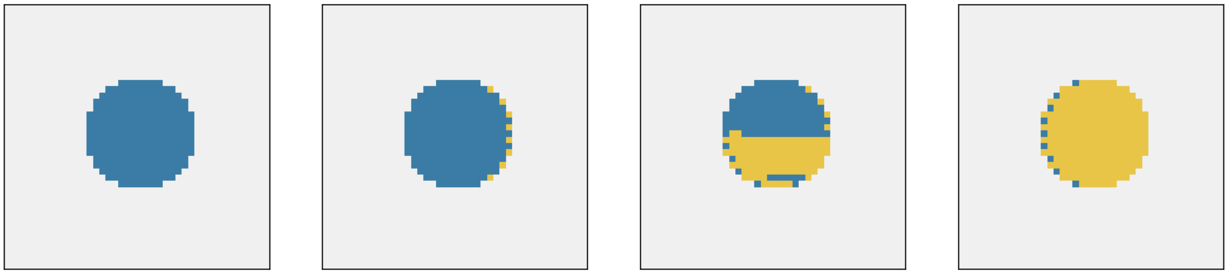
A deterministic trajectory stresses every die almost identically, so sweeping its parameters mainly moves the *count* of failing dies (Experiment 3). Spatially structured, mechanism-bearing patterns can only arise where dies genuinely experience different dynamics. This experiment isolates the strongest such mechanism in the step-and-scan cycle: the settling budget t_{step} between motions. Exposure of each die begins immediately after the preceding step, so any residual transient at scan entry is part of the exposed error; dies whose preceding step is atypical — above all the row-change dies at the wafer edge, whose repositioning move spans the difference in row length on a circular wafer — enter the scan with a larger residual. As t_{step} shrinks (raising throughput), those dies should fail first, producing a scanner-plausible edge pattern that emerges from the motion profile itself, with no injected disturbances. Demonstrating this is demonstrating the central premise of the thesis: that trajectory aggressiveness maps to defect morphology, not merely to an error norm.

The experiment runs the baseline trajectory under Case 3b control on the 241-die wafer, sweeping t_{step} from 200 to 120 ms at a fixed failure threshold (twice the median exposure MSD of the 200 ms run, 1.37 µm on the corrected plant). Two spatial statistics are tracked besides the failing count: the *row-change enrichment* (the failure rate among post-row-change dies relative to the wafer average) and the *scan-order index* (mean scan position of the failing dies, normalized so 1.0 is uniform and values below 1 indicate that early-scanned dies fail).

Table 5.6.5: Throughput–yield transition under decreasing settling budget (baseline trajectory, Case 3b, 241 dies).

t_{step}	Failing dies	Row-change enrichment	Scan-order index	CNN class	$\mathcal{J}_{map}$	T_{wafer}
200 ms	0/241	—	—	none (97.2 %)	0.041	78.3 s
190 ms	0/241	—	—	none (97.2 %)	0.041	75.9 s
180 ms	8/241	**15.1**	0.94	none (97.4 %)	0.037	73.5 s
170 ms	8/241	15.1	0.94	none (97.4 %)	0.037	71.1 s
160 ms	8/241	15.1	0.94	none (97.4 %)	0.037	68.7 s
150 ms	107/241	1.13	0.54	Loc (75.2 %)	1.999	66.3 s
120 ms	232/241	0.52	1.00	Center (100 %)	2.000	59.0 s

The results (Table 5.6.5, Figure 5.10) confirm the mechanism precisely at the transition onset. At $t_{step} = 180$ ms the first eight failures appear with a row-change enrichment of **15.1**: every one of them is a die scanned immediately after a row change, clustering along one edge of the wafer — visually an *Edge-Loc* morphology in the WM-811K taxonomy, produced purely by the motion profile. The onset is a plateau, not a knife edge: the same eight-die cluster holds unchanged from 180 down to 160 ms, and the wafer-wide collapse occurs between 160 and 150 ms. Deeper into the transition the structure changes character: at 150 ms nearly half the wafer fails in a band covering the *early-scanned* rows (scan-order index 0.54), a fingerprint of a slowly decaying startup transient interacting with the shortened settling budget; by 120 ms failure is near-total. The wafer cycle time falls from 78.3 to 59.0 s across the sweep, making the throughput–yield



(a) 200 ms (b) 180 ms (c) 150 ms (d) 120 ms

Figure 5.10: Wafer maps across the throughput–yield transition. At 180 ms the first failures form an edge cluster at the row-change dies; at 150 ms a scan-order-correlated band covers the early-scanned half of the wafer; by 120 ms failure is near-total.

trade explicit: between 190 and 150 ms of settling budget, a 13 % cycle-time gain costs the wafer 44 % of its dies.

Two observations qualify the classifier's part. The eight-die edge cluster at the onset is below the CNN's sensitivity (it still reports *none* at 97.4 %, and the cost even dips slightly — a reminder that the cost is a probability, not a die count); the mid-transition band at 150 ms is read as *Loc*, and the saturated maps beyond it as *Center* — classes that the corrected cost (4.5.2) prices at its maximum as out-of-domain signals instead of at zero. The informative segment of the cost curve — exactly as anticipated — is the transition region, and it is there that the optimizer of Experiment 6 operates.

5.7 Experiment 6: Optimization Scenarios E3 and E4

The final experiment runs the particle swarm of Section 4.5 twice over the same plant, controller (Case 3b), settling budget ($t_{step} = 150$ ms, inside the transition region), threshold, and throughput constraint ($T_{budget} = 15$ s per 49-die wafer, $\beta = 4$), changing only the cost: scenario E3 minimizes the conventional servo cost (wafer-average exposure MSD normalized by the threshold), and scenario E4 minimizes the yield-aware CNN cost (4.5.2). The search runs over the exact fifteen-segment profile family, so all four bounds — including $s_{\max}$ — are live dimensions. Six particles run for eight iterations (48 full-wafer simulations per scenario), with one particle seeded at the baseline trajectory.

Table 5.7.6: Optimization scenarios under identical constraints, exact snap-bounded generator. Failing dies and $\mathcal{J}_{map}$ are evaluated on the optimizer's own best candidate.

Scenario	v [m/s]	a [m/s^2]	j [m/s^3]	s [m/s^4]	Failing	CNN class	$\mathcal{J}_{map}$	T_{wafer}
Baseline (seed)	0.80	20.0	1600	10^5	4/49	none	0.038	15.2 s
E3: servo cost	1.50	40.5	1512	5×10^5	0/49	none	0.043	14.8 s
E4: CNN cost	1.22	38.6	2589	3.9×10^5	3/49	none	0.026	13.5 s

Both searches converge, and both dominate the baseline seed — which, under the exact generator's longer scan profile, slightly exceeds the throughput budget and carries four failing dies (Table 5.7.6). The servo cost drives the velocity to the *upper bound of the search box*: on the frictionless plant, raising v both shortens the point-exposure window (lowering every die's MSD, the reversal documented in Experiment 3) and shortens the scan, so no interior trade-off remains within the admissible box and the search settles at $v = 1.5\,\mathrm{m/s}$ with the throughput budget inactive. Its reward mechanism is worth scrutiny — a shorter integration window lowers MSD for reasons whose photolithographic side effects (dose, imaging) lie outside the model.

The CNN cost run exposes the cost's calibration properties more sharply than any previous experiment: its returned optimum is *not clean*. A map with three failing dies scores $\mathcal{J}_{map} = 0.026$, below the 0.043 of a perfectly clean map and the 0.038 of the four-die baseline map, so the swarm actively converges onto slightly-imperfect wafers — real WM-811K wafers labelled *none* typically carry a few scattered failing dies, and the network scores similarity to that training distribution rather than the die count. On the uncorrected plant this preference merely produced flatness across clean maps; on the corrected plant, where the optimizer can reach many clean candidates, it inverts the ranking at the top of the search. The E4 optimum is 9 % faster than the servo optimum (13.5 vs. 14.8 s) and trades three dies for that speed — a trade a calibrated yield cost would have to price explicitly rather than reward. Both observations point to the same refinement: probability calibration of P_{none} plus a small die-count regularizer, giving the cost a slope on the clean plateau and aligning it with monotone yield. With that pass, the demonstrated machinery — swarm, exact profile family, full-wafer evaluation, CNN pricing — is positioned to evaluate the dominance claim of hypothesis H3 on the yield–throughput plane. The cost-evaluation history of both runs is archived with the experiment scripts.

5.8 Experiment 7: Does Trajectory Smoothness Matter Without a Good Controller?

The premise of higher-order trajectory design is that bounding deeper derivatives injects less energy into the structure. A fair question is whether that refinement survives contact with a mediocre controller: if the tracking error is dominated by servo lag, does smoothing the reference accomplish anything at all? This experiment crosses the profile order — acceleration-bounded (second order), jerk-bounded (third), and snap-bounded (fourth, exact fifteen-segment generator, plus a tight-snap variant at $s_{\max} = 2 \times 10^4\,\mathrm{m/s^4}$) — with the two ends of the controller ladder: bare collocated PID (Case 1) and feedforward plus fine-stage HIGS (Case 3b). All profiles run at the same kinematic bounds $(0.8, 20, 1600)$ on the four-die row.

The answer is layered (Table 5.8.7, Figure 5.11). **Under the bare PID, trajectory order reaches the exposure only through the transients.** The cruise MA sits at 0.75–1.0 mm for every order — it is the slowly decaying tail of the acceleration transient, which the collocated loop cannot reject, and reference smoothness does not remove it within the cruise window. Order

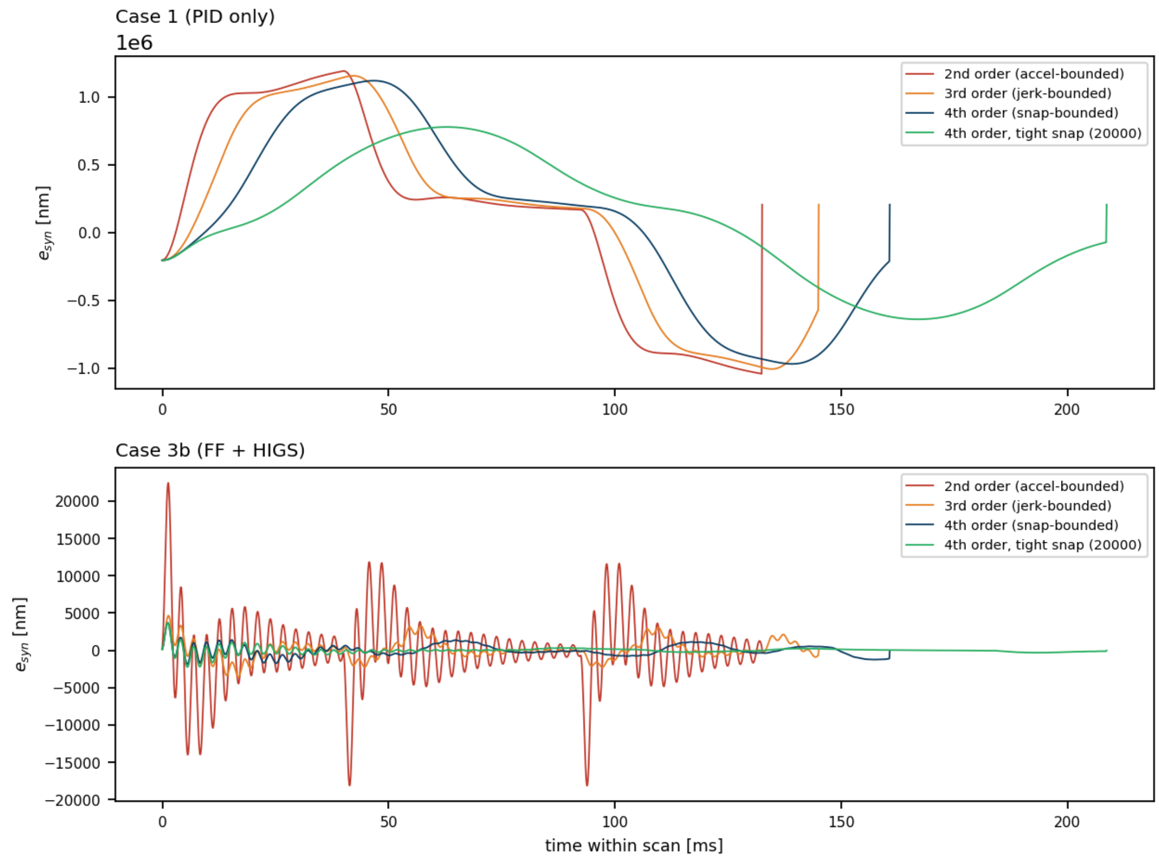


Figure 5.11: Synchronization error during one scan for the four profile orders, under bare PID (top) and under the full controller ladder (bottom). Note the vertical scales.

Table 5.8.7: Profile order × controller. Cruise indices of e_{syn} (worst die); full-scan MSD includes the post-step entry transient.

Profile	Controller	T_{scan}	cruise max|MA|	cruise max MSD	full-scan MSD
2nd order	Case 1	132.5 ms	834 µm	54.9 µm	293 µm
3rd order	Case 1	145.0 ms	946 µm	36.1 µm	198 µm
4th order	Case 1	160.8 ms	1001 µm	26.3 µm	105 µm
4th, tight snap	Case 1	208.6 ms	749 µm	12.5 µm	67 µm
2nd order	Case 3b	132.5 ms	8.38 µm	6.84 µm	10.6 µm
3rd order	Case 3b	145.0 ms	2.94 µm	0.90 µm	1.6 µm
4th order	Case 3b	160.8 ms	1.33 µm	0.18 µm	1.7 µm
4th, tight snap	Case 3b	208.6 ms	**0.33** µm	**0.046** µm	1.8 µm

does shape how much transient there is to decay: the cruise MSD improves monotonically with smoothness, 54.9 → 36.1 → 26.3 → 12.5 µm from second order to tight snap (4.4×), and the full-scan MSD follows the same ordering (293 → 67 µm); the reduction is real but the exposure error remains dominated by what the controller leaves behind.

Under the full ladder, trajectory order is worth more than an order of magnitude. With the feedforward removing the quasi-static lag and the HIGS loop suppressing what lies within its bandwidth, the cruise MSD falls 6.84 → 0.90 → 0.18 µm from second to fourth order — a 38× improvement attributable to the profile alone, at identical kinematic bounds. The interpretation is classical and now quantified for this plant: the fine loop handles the error components inside its bandwidth, and the profile order determines how much excitation exists *beyond* it; smoothing the reference and improving the controller are complements, not substitutes. Two further readings matter for the optimizer. First, smoothness has a throughput price — T_{scan} grows 21 % from second to fourth order at the same bounds, and a further 30 % for the tight-snap variant — so profile order participates in the same throughput–yield trade as the kinematic bounds themselves. Second, the tight-snap variant is now the *most* accurate configuration (cruise MSD 46 nm, MA 0.33 µm — the closest any configuration of this plant comes to the nanometre specifications), but its full-scan MSD is no better than the moderate profile's (1.8 vs. 1.7 µm, both set by the post-step entry transient) and it spends 30 % more scan time. Accuracy inside the cruise keeps improving with smoothness, while the settling-dominated component saturates at the closed loop's floor and the throughput cost grows without bound: the optimum in the snap dimension is therefore interior once throughput enters the cost — precisely the trade the optimizer of Experiment 6 is built to resolve.

5.9 A Recurrent Surrogate Model (Secondary Study)

This closing study is deliberately *secondary*: the main line of the thesis — trajectory generation, control, defect mapping, CNN cost, and optimization — does not depend on it, and Experiment 6 runs its optimization against the full simulation directly. The recurrent surrogate of Section 4.7, trained on 100 randomized simulation runs ($\approx$ 1.6 million samples), is evaluated in two roles: as a fast predictor of the exposure indices for individual candidate trajectories, and as the inner evaluation of a grid search over the kinematic bounds.

Table 5.9.8 shows the surrogate's predictions for three trajectories of decreasing aggressiveness, after retraining on the corrected plant. The predicted indices decrease monotonically — by a factor of 30 on MA and 45 on MSD between the baseline and the slowest candidate — reproducing the ordering observed in the full simulation. Their absolute scale, however, inherits the constant geometric offset present in the synchronization-error signal on which the surrogate was trained, so the predictions are meaningful for *ranking* candidates rather than for verifying absolute compliance; all three candidates are correctly rejected against the nanometre thresholds.

The grid search evaluates 450 configurations over $v \in [0.01, 0.5]\,\mathrm{m/s}$, $a \in [0.1, 10]\,\mathrm{m/s^2}$, $j \in \{500, 1000, 2000\}\,\mathrm{m/s^3}$, $s \in \{10^4, 5 \times 10^4, 10^5\}\,\mathrm{m/s^4}$, accepting a candidate only if the predicted

Table 5.9.8: Surrogate predictions of the exposure indices for three candidate trajectories.

$v_{\max}$ [m/s]	$a_{\max}$ [m/s^2]	$j_{\max}$ [m/s^3]	$s_{\max}$ [m/s^4]	$\max\lvert\mathrm{MA}\rvert$	$\max\mathrm{MSD}$	Verdict
0.80	20.0	1600	10^5	31.5 mm	5.26 mm	FAIL
0.10	1.0	100	10^4	2.28 mm	0.46 mm	FAIL
0.01	0.1	1	10	1.09 mm	0.118 mm	FAIL

indices satisfy the nanometre specifications. The search terminates without finding any feasible candidate and reports that the sub-nanometre regime is unreachable for the Case 1 machine, recommending the spring-compensation or fine-stage configurations of the reference study. This negative result is consistent with Experiment 1 and with the sweep of Experiment 3: in a coarse-only machine, no admissible trajectory can suppress the reticle-coupling error to nanometre level, since the dominant error term is structural rather than kinematic.

5.9.1 Is a recurrent surrogate the right tool?

The results above prompt a question that the thesis should answer explicitly rather than assume: does a learned sequence model make sense for this plant at all?

For the Case 1 configuration as simulated, the honest answer is that it is not *necessary*. The plant is linear time-invariant — linear Kelvin–Voigt couplings, linear PID feedback, linear feedforward — so the map from the reference kinematics to the synchronization error is an exact, known LTI operator. It can be evaluated by directly integrating the reduced two-body model (seconds per candidate, Experiment 1) or captured without error by a fitted low-order transfer function, with no training data and full interpretability. A GRU trained on one hundred randomized runs approximates something that, in this regime, can be computed exactly; its only measured advantage is evaluation speed against the *full* MuJoCo model, not against the reduced model that suffices for Case 1. The training data also transferred a defect into the surrogate:

the synchronization-error signal it learned from contains the constant geometric offset of the absolute coordinates, which is why the predictions of Table 5.9.8 rank correctly but carry a meaningless absolute scale.

The recurrent surrogate earns its place under three conditions that define the continuation of this work, none of which holds in the linear baseline but all of which appear in the target configurations. First, *nonlinearity*: the actuator saturation (5×10^4 N) engaged by aggressive transients, and above all the nonlinear feedback structures planned for the fine-stage configurations (reset elements and hybrid integrator–gain systems[6]), break the LTI shortcut — there is then no exact transfer function to fit, and a learned dynamic model becomes a legitimate system-identification tool. Second, *cost amortization*: when the evaluation plant is the full multi-body simulation over a complete 241-die wafer (minutes per candidate, Experiment 3) inside a population-based optimizer issuing thousands of evaluations, a millisecond surrogate changes what is computationally feasible. Third, *inter-segment memory*: the recurrent state carries transients across the scan–step–scan sequence, an effect that static regressions over $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ cannot represent and that grows in importance as settling budgets shrink.

Three concrete amendments follow for the next iteration of the surrogate: retrain on the offset-corrected error signal, so that absolute predictions become meaningful against the MA/MSD thresholds; report ranking fidelity (for example rank correlation against full-simulation sweeps) as the primary metric, since ranking is what the optimizer consumes; and benchmark against the LTI baseline on Case 1, so that the surrogate's added value is demonstrated where it exists — the nonlinear, full-model configurations — rather than presumed everywhere.

5.10 Discussion

Read against the evaluation scenarios of the thesis proposal — E1: third-order baseline, E2: fourth-order baseline, E3: optimization under a servo cost, E4: optimization under the CNN cost — the experiments above complete E2, E3, and E4: the trajectory generator, the validated plant, the controller ladder up to the nonlinear HIGS loop, the trained classifier, the trajectory-to-map bridge, and two full optimization runs differing only in their cost. E1 remains open, and deliberately so: a fair third-versus-fourth-order comparison requires the exact fifteen-segment snap-limited generator (Section 4.3) so that both profile orders are represented by their true shapes rather than by polynomial approximations.

Of the validation criterion, three elements are now satisfied: unperturbed operation maps to the *none* class; the cost input shifts monotonically and reproducibly under parameter sweeps; and — the strongest result of the campaign — spatially structured, scanner-plausible morphology emerges from the motion profile itself, with the transition onset of Experiment 5 concentrating its first failures at the row-change dies with fifteen-fold enrichment, an *Edge-Loc* pattern requiring no injected disturbance.

The class-level half of the criterion — that motion-induced failures be *classified* into the scanner-plausible classes — is met only partially, and every failure mode observed points at the classifier rather than at the simulation half of the pipeline. The scratch-shaped disturbance cannot be recognized as *Scratch* while the classifier's recall for that class is zero (Experiment 2), so the 15-die column falls to *Loc*; dense scattered patterns are absorbed into *Center*; the drift annulus is read as *Donut*; the eight-die Edge-Loc onset of Experiment 5 sits below the classifier's sensitivity; and Experiment 6 showed the optimizer converging onto a slightly-imperfect wafer because the classifier scores it as more *none*-like than a clean one. The refinement list is correspondingly concrete and bounded: rebalanced training for the minority classes, probability calibration on clean and near-clean maps, and a die-count regularizer in the cost. On the trajectory side the exact generators are now in place, and Experiment 7 showed that profile order is worth well over an order of magnitude of exposure error under a capable controller, with the snap trade-off resolved by throughput rather than by accuracy alone. On the plant side, the corrections adopted in this iteration — error-velocity servo damping on the frictionless guideways, balance-mass cancellation of the actuator reactions, and consistently paired error sampling — brought the exposure error from the millimetre scale to the sub-micrometre scale; the remaining three orders of magnitude to the nanometre specifications belong to the fine positioning stage, which in the reference machine is a purpose-designed 2 kHz subsystem rather than the bandwidth-limited generic servo modelled here. None of these items changes the architecture validated here: generate, track, map, classify, price, and optimize — the loop is closed and runs end to end.

Conclusion

This thesis set out to connect two worlds that the wafer-scanner literature treats separately: the servo-mechanical design of reference trajectories, evaluated through settling times and filtered error indices, and the yield engineering of the fab, expressed in the morphology of defect patterns on the wafer. The inverse direction — diagnosing a process cause from an observed wafer fingerprint — is established industrial practice; the forward direction — predicting, from a set of kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$, the defect pattern they will produce, and optimizing against it — is the gap this work addresses.

The infrastructure for that forward direction was built and validated in simulation. A planar multi-body model of a generic lithography machine, organized as reticle, optics, and wafer chains coupled through Kelvin–Voigt elements, was implemented both as a full MuJoCo simulation and as a reduced two-body-per-chain representation. The reduced model reproduces the published Case 1 study it derives from: a machine without fine stages, under collocated PID control with inertia feedforward, misses the exposure specifications by nearly three orders of magnitude (MA within $\pm 985\,\mathrm{nm}$ against a $[-1.25, +2]\,\mathrm{nm}$ band), with the error dominated by the lightly damped $131\,\mathrm{Hz}$ resonance of the reticle coupling excited during the acceleration ramps.

On the classification side, a convolutional network trained on the labelled portion of the industrial WM-811K dataset (172,950 wafer maps) reached $96.88\,\%$ hold-out accuracy with a strongly diagonal confusion matrix. Its known weakness — zero recall on the *Scratch* class, a direct consequence of class imbalance — was characterized and its mitigation deferred to the training recipe, where it belongs.

The experiments connecting the two halves produced six concrete findings. First, the exposure-window error responds reproducibly to every active kinematic bound, monotonically in the force-shaping ones: across the sweep, the median exposure MSD scales from 45–52 μm (conservative, quarter jerk) through 152 μm (baseline) to 388 μm (aggressive) — while lowering the scan velocity *raises* the per-die error, because the point-exposure window scales as $1/v$ and integrates more of the decaying transient. Second, the control ladder built on top of the baseline — identified lag feedforward, then a fine-stage loop closed on the measured end-effector error — reduces the exposure error by more than two orders of magnitude (33.9 to 0.22 μm of MSD, 738 to 2.1 μm of MA), with the linear PI and the nonlinear HIGS integrator performing within 25 % of each other at their stability-boundary tunings. Third, and central to the thesis premise: as the settling budget shrinks, the first trajectory-induced failures appear at the row-change dies with fifteen-fold enrichment — an edge-local, scanner-plausible pattern produced by the motion profile alone, stable across a 20 ms plateau of settling budgets — before a scan-order-correlated band consumes the wafer; the informative segment of the cost lives in this transition region. Fourth, trajectory smoothness and controller quality are complements, not substitutes: under a bare PID, profile order reduces transients (4.4× on exposure MSD from second order to tight snap) but cannot reach the exposure window through the dominant transient tail, whereas under the full controller ladder the same change of profile order is worth 38× on the exposure MSD at identical kinematic bounds — and while cruise accuracy keeps improving with tighter snap (down to 46 nm of MSD), the settling-dominated error saturates and the scan time grows, so the optimum in the snap dimension is interior once throughput enters the cost. Fifth, the particle-swarm scenarios closed the loop end to end over the exact snap-bounded profile family: both costs dominate the baseline seed, with the servo cost driving velocity to the search bound and the CNN cost reaching its optimum 9 % faster still — while exposing, through the optimizer itself, the calibration property the cost must fix before a dominance claim can be made: a learned preference for slightly-imperfect wafers over perfect ones, inherited from the labelling statistics of real production data, which on this plant inverts the ranking at the top of the search and returns a three-failing-die wafer as the optimum. Sixth, on the way to these results the original polynomial generator was found to bound snap only implicitly; the exact fifteen-segment generator (with second- and third-order counterparts) was implemented and verified, restoring $s_{\max}$ as a live dimension of the search space.

The validation criterion for the trajectory-to-defect mapping was met in its spatial and reproducibility halves, and partially in its classification half: unperturbed operation classifies as *none*, the cost shifts reproducibly under parameter sweeps, and structured plausible morphology emerges from the trajectory itself; but the assignment of those patterns to the intended minority classes remains limited by the classifier's scratch recall, sensitivity, and calibration. All observed shortfalls localize in the classifier and cost calibration — bounded refinements — rather than in the forward architecture, which runs end to end: generate, track, map, classify, price, optimize.

The recurrent surrogate model was evaluated critically as well as empirically. Its predictions rank candidate trajectories correctly but inherit the geometric offset of their training signal, and — more fundamentally — for the linear Case 1 plant a classical LTI evaluation would be exact and data-free. The surrogate is therefore retained not as a replacement for analysis where analysis suffices, but for the regimes the framework is headed toward: saturated actuation, nonlinear feedback structures, and full-wafer simulations whose cost dominates a population-based search. Retraining on the offset-corrected error and benchmarking against the linear baseline are prerequisites for that role.

The immediate continuation of this work follows directly from the findings: retrain the classifier with class rebalancing and probability calibration, add a die-count regularizer to the cost, and re-run the optimization scenarios on the calibrated cost to evaluate the dominance claim of hypothesis H3 — now over the full yield-versus-throughput Pareto front, since Experiment 7 established that profile order itself participates in that trade. Beyond these, two extensions raise the fidelity ceiling: a fine positioning stage representative of the reference machine's purpose-designed high-bandwidth subsystem (the generic fine-stage servo modelled here caps its authority at $\sim$100 rad/s and leaves the last three orders of magnitude to the nanometre specifications unclosed, even after the plant corrections — frictionless guideways and balance-mass reaction cancellation — adopted in this work brought the exposure error from millimetres to sub-micrometres), and a deeper study of the HIGS loop in the overshoot-constrained settling problems where its phase advantage over linear integrators is expected to materialize.

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Appendix A

Full-State Space Model

A.1 State-Space Equations for the Wafer Chain

The wafer chain corresponds to $j = 3$, with $N_3 = 7$ (bodies $i = 2, \ldots, 7$). These map to global state vectors $\mathbf{x}_{2p-1}$ (translational) and $\mathbf{x}_{2p}$ (rotational) for $p = 12, \ldots, 17$. The state-space equations are represented below.

First body of the wafer chain ($i = 2$, $p = 12$):

$$\begin{aligned}\ddot{\mathbf{x}}_{23} &= \frac{1}{\bar{m}_{2_3}}\Big[u^0_{2_3} - {}^0\bar{C}_{2_3}\dot{\mathbf{x}}_{23} - {}^0\bar{K}_{2_3}\mathbf{x}_{23} + {}^0\hat{C}_{3_3}\dot{\mathbf{x}}_{25} + {}^0\hat{K}_{3_3}\mathbf{x}_{25}\Big] \\ &\quad - \frac{1}{m_0}\Big[u^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_1 - {}^0\bar{K}_1\mathbf{x}_1 + \sum_{k=1}^{3}\Big({}^0\hat{C}_{2_k}\dot{\mathbf{x}}_{2p_{2_k}-1} + {}^0\hat{K}_{2_k}\mathbf{x}_{2p_{2_k}-1}\Big)\Big]\end{aligned} \tag{A.1.1}$$

$$\begin{aligned}\ddot{\mathbf{x}}_{24} &= \frac{1}{I_{z_{2_3}}}\Big[\tau^0_{2_3} - {}^0\bar{c}_{\theta,2_3}\dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,2_3}\mathbf{x}_{24} + {}^0\hat{c}_{\theta,3_3}\dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,3_3}\mathbf{x}_{26}\Big] \\ &\quad - \frac{1}{I_{z_0}}\Big[\tau^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_2 - {}^0\bar{K}_1\mathbf{x}_2 + \sum_{k=1}^{3}\Big({}^0\hat{c}_{\theta,2_k}\dot{\mathbf{x}}_{2p_{2_k}} + {}^0\hat{k}_{\theta,2_k}\mathbf{x}_{2p_{2_k}}\Big)\Big]\end{aligned} \tag{A.1.2}$$

Intermediate body ($i = 3$, $p = 13$):

$$\begin{aligned}\ddot{\mathbf{x}}_{25} &= \frac{1}{\bar{m}_{3_3}}\Big[u^0_{3_3} - {}^0\bar{C}_{3_3}\dot{\mathbf{x}}_{25} - {}^0\bar{K}_{3_3}\mathbf{x}_{25} + {}^0\hat{C}_{(4)_3}\dot{\mathbf{x}}_{27} + {}^0\hat{K}_{(4)_3}\mathbf{x}_{27}\Big] \\ &\quad - \frac{1}{\bar{m}_{(2)_3}}\Big[u^0_{(2)_3} - {}^0\bar{C}_{(2)_3}\dot{\mathbf{x}}_{23} - {}^0\bar{K}_{(2)_3}\mathbf{x}_{23} + {}^0\hat{C}_{3_3}\dot{\mathbf{x}}_{25} + {}^0\hat{K}_{3_3}\mathbf{x}_{25}\Big]\end{aligned} \tag{A.1.3}$$

$$\begin{aligned}\ddot{\mathbf{x}}_{26} &= \frac{1}{I_{z_{3_3}}}\Big[\tau^0_{3_3} - {}^0\bar{c}_{\theta,3_3}\dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,3_3}\mathbf{x}_{26} + {}^0\hat{c}_{\theta,(4)_3}\dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,(4)_3}\mathbf{x}_{28}\Big] \\ &\quad - \frac{1}{I_{z_{(2)_3}}}\Big[\tau^0_{(2)_3} - {}^0\bar{c}_{\theta,(2)_3}\dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,(2)_3}\mathbf{x}_{24} + {}^0\hat{c}_{\theta,3_3}\dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,3_3}\mathbf{x}_{26}\Big]\end{aligned} \tag{A.1.4}$$

Intermediate body $(i = 4,\ p = 14)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{27} &= \frac{1}{\bar{m}_{4_3}}\Big[u^0_{4_3} - {}^0\bar{C}_{4_3}\dot{\mathbf{x}}_{27} - {}^0\bar{K}_{4_3}\mathbf{x}_{27} + {}^0\hat{C}_{(5)_3}\dot{\mathbf{x}}_{29} + {}^0\hat{K}_{(5)_3}\mathbf{x}_{29}\Big] \\
&\quad - \frac{1}{\bar{m}_{(3)_3}}\Big[u^0_{(3)_3} - {}^0\bar{C}_{(3)_3}\dot{\mathbf{x}}_{25} - {}^0\bar{K}_{(3)_3}\mathbf{x}_{25} + {}^0\hat{C}_{4_3}\dot{\mathbf{x}}_{27} + {}^0\hat{K}_{4_3}\mathbf{x}_{27}\Big] \qquad \text{(A.1.5)} \\
\ddot{\mathbf{x}}_{28} &= \frac{1}{I_{z_{4_3}}}\Big[\tau^0_{4_3} - {}^0\bar{c}_{\theta,4_3}\dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,4_3}\mathbf{x}_{28} + {}^0\hat{c}_{\theta,(5)_3}\dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,(5)_3}\mathbf{x}_{30}\Big] \\
&\quad - \frac{1}{I_{z_{(3)_3}}}\Big[\tau^0_{(3)_3} - {}^0\bar{c}_{\theta,(3)_3}\dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,(3)_3}\mathbf{x}_{26} + {}^0\hat{c}_{\theta,4_3}\dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,4_3}\mathbf{x}_{28}\Big] \qquad \text{(A.1.6)}
\end{aligned}$$

Intermediate body $(i = 5,\ p = 15)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{29} &= \frac{1}{\bar{m}_{5_3}}\Big[u^0_{5_3} - {}^0\bar{C}_{5_3}\dot{\mathbf{x}}_{29} - {}^0\bar{K}_{5_3}\mathbf{x}_{29} + {}^0\hat{C}_{(6)_3}\dot{\mathbf{x}}_{31} + {}^0\hat{K}_{(6)_3}\mathbf{x}_{31}\Big] \\
&\quad - \frac{1}{\bar{m}_{(4)_3}}\Big[u^0_{(4)_3} - {}^0\bar{C}_{(4)_3}\dot{\mathbf{x}}_{27} - {}^0\bar{K}_{(4)_3}\mathbf{x}_{27} + {}^0\hat{C}_{5_3}\dot{\mathbf{x}}_{29} + {}^0\hat{K}_{5_3}\mathbf{x}_{29}\Big] \qquad \text{(A.1.7)} \\
\ddot{\mathbf{x}}_{30} &= \frac{1}{I_{z_{5_3}}}\Big[\tau^0_{5_3} - {}^0\bar{c}_{\theta,5_3}\dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,5_3}\mathbf{x}_{30} + {}^0\hat{c}_{\theta,(6)_3}\dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,(6)_3}\mathbf{x}_{32}\Big] \\
&\quad - \frac{1}{I_{z_{(4)_3}}}\Big[\tau^0_{(4)_3} - {}^0\bar{c}_{\theta,(4)_3}\dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,(4)_3}\mathbf{x}_{28} + {}^0\hat{c}_{\theta,5_3}\dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,5_3}\mathbf{x}_{30}\Big] \qquad \text{(A.1.8)}
\end{aligned}$$

Intermediate body $(i = 6,\ p = 16)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{31} &= \frac{1}{\bar{m}_{6_3}}\Big[u^0_{6_3} - {}^0\bar{C}_{6_3}\dot{\mathbf{x}}_{31} - {}^0\bar{K}_{6_3}\mathbf{x}_{31} + {}^0\hat{C}_{(7)_3}\dot{\mathbf{x}}_{33} + {}^0\hat{K}_{(7)_3}\mathbf{x}_{33}\Big] \\
&\quad - \frac{1}{\bar{m}_{(5)_3}}\Big[u^0_{(5)_3} - {}^0\bar{C}_{(5)_3}\dot{\mathbf{x}}_{29} - {}^0\bar{K}_{(5)_3}\mathbf{x}_{29} + {}^0\hat{C}_{6_3}\dot{\mathbf{x}}_{31} + {}^0\hat{K}_{6_3}\mathbf{x}_{31}\Big] \qquad \text{(A.1.9)} \\
\ddot{\mathbf{x}}_{32} &= \frac{1}{I_{z_{6_3}}}\Big[\tau^0_{6_3} - {}^0\bar{c}_{\theta,6_3}\dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,6_3}\mathbf{x}_{32} + {}^0\hat{c}_{\theta,(7)_3}\dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,(7)_3}\mathbf{x}_{34}\Big] \\
&\quad - \frac{1}{I_{z_{(5)_3}}}\Big[\tau^0_{(5)_3} - {}^0\bar{c}_{\theta,(5)_3}\dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,(5)_3}\mathbf{x}_{30} + {}^0\hat{c}_{\theta,6_3}\dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,6_3}\mathbf{x}_{32}\Big] \qquad \text{(A.1.10)}
\end{aligned}$$

End-effector body $(i = 7,\ p = 17)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{33} &= \frac{1}{\bar{m}_{7_3}}\Big[u^0_{7_3} - {}^0\bar{C}_{7_3}\dot{\mathbf{x}}_{33} - {}^0\bar{K}_{7_3}\mathbf{x}_{33}\Big] \\
&\quad - \frac{1}{\bar{m}_{(6)_3}}\Big[u^0_{(6)_3} - {}^0\bar{C}_{(6)_3}\dot{\mathbf{x}}_{31} - {}^0\bar{K}_{(6)_3}\mathbf{x}_{31} + {}^0\hat{C}_{7_3}\dot{\mathbf{x}}_{33} + {}^0\hat{K}_{7_3}\mathbf{x}_{33}\Big] \qquad \text{(A.1.11)} \\
\ddot{\mathbf{x}}_{34} &= \frac{1}{I_{z_{7_3}}}\Big[\tau^0_{7_3} - {}^0\bar{c}_{\theta,7_3}\dot{\mathbf{x}}_{34} - {}^0\bar{k}_{\theta,7_3}\mathbf{x}_{34}\Big] \\
&\quad - \frac{1}{I_{z_{(6)_3}}}\Big[\tau^0_{(6)_3} - {}^0\bar{c}_{\theta,(6)_3}\dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,(6)_3}\mathbf{x}_{32} + {}^0\hat{c}_{\theta,7_3}\dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,7_3}\mathbf{x}_{34}\Big] \qquad \text{(A.1.12)}
\end{aligned}$$